

# **LABORATORY 07 - BASIC INFRASTRUCTURE AND NETWORK LAYER**

Laboratory 7 corresponding to the third term

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**NETWORK ARCHITECTURE AND SERVICES**

**GROUP 3**

Bogotá DC, Colombia

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# **Lab No.07 - BASIC INFRASTRUCTURE AND NETWORK LAYER**

## **Objectives**

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- Continue learning about the operation of operating systems and network services.
- Install network management tools.
- Configure routers and static routing.

## **Tools to be used**

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- Computers
- Internet access
- Virtualization software
- Packet Tracer and Wireshark
- Routers and switches

## **Introduction**

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Modern organizations rely heavily on robust and well-managed network infrastructures to ensure the efficient operation of their technological platforms. These infrastructures typically integrate wired and wireless user stations, physical and virtualized servers, switches, routers, and cloud-based services. Proper configuration and monitoring of these components are essential for guaranteeing availability, performance, and security across the entire system.

This laboratory aims to deepen the understanding of how operating systems, network devices, and management tools operate within such environments. Students will explore network-related commands across different operating systems, implement monitoring services, manage cloud-based resources through Microsoft Azure, and analyze network paths using ICMP-based tools. Additionally, the lab introduces hands-on experience with physical routers, including console access, basic configuration, serial interconnection, and the implementation of static routing.

By completing this laboratory, students will strengthen their ability to manage real networking environments, interpret diagnostic outputs, and configure routing and monitoring tools essential for modern corporate networks.

## Theoretical Framework

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A modern network infrastructure brings together operating systems, communication protocols, and interconnected devices to ensure efficient and secure organizational operations. Operating systems provide essential diagnostic tools such as *netstat*, *vnstat*, *route*, and *ethtool*, which allow administrators to examine active connections, interface performance, and routing behavior. Mastering these tools is crucial for troubleshooting, optimizing system performance, and understanding how data flows across a network.

Network monitoring is another key aspect of infrastructure management. Through protocols like SNMP (Simple Network Management Protocol), administrators can remotely collect information on CPU usage, memory, disk space, and network activity. SNMP's manager-agent model enables real-time visibility of system health, helping prevent failures and ensuring reliable service delivery. Cloud platforms such as Microsoft Azure extend this capability by offering integrated telemetry services like Application Insights, which track request rates, response times, failures, and user activity across geographically distributed environments.

Diagnostic tools based on the Internet Control Message Protocol (ICMP), such as *ping* and *traceroute*, provide valuable insights into connectivity and path analysis. These tools reveal latency, packet loss, and the route that packets follow across the Internet, making them essential for identifying bottlenecks and evaluating network behavior.

Routers play a central role in forwarding packets between networks. They rely on different memory types—ROM, RAM, NVRAM, and Flash—to store the operating system, startup configurations, and routing information. Initial access to routers is commonly performed through a console connection, where administrators configure passwords, hostnames, interface descriptions, and other basic parameters necessary for secure and functional operation.

Serial interconnections between routers simulate wide-area network links in controlled environments. These links distinguish between DCE (Data Communications Equipment) and DTE (Data Terminal Equipment), with the DCE device providing the required clock rate for synchronization. Once physical connectivity is established, static routing allows administrators to manually define paths to remote networks. Although simple, static routing is highly effective in laboratory conditions where topology changes are predictable. Using *ping* and *traceroute*, students can verify correct routing and confirm end-to-end connectivity.

# Base Software Installation

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In the same groups as throughout this laboratory, perform the following activities.

## 1. Other Useful Commands

a. Study the functionality of commands that provide network information, e.g., netstat, vnstat, route, and ethtool (or equivalents in Slackware, Solaris, Centos, and PowerShell). Review different parameters that can be used and create a Shell program that uses them (create a menu with at least 5 different options that show different executions of the commands). Students should be able to understand the output of the commands and present it in an easy-to-understand format.

On Linux Slackware

```
==== Network Information Utility =====
1) Mostrar interfaces de red
2) Mostrar tabla de rutas
3) Mostrar conexiones activas
4) Mostrar estadísticas de red
5) Mostrar información de una interfaz
0) Salir
Seleccione una opción:
1
---- Network Interfaces ----
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
   link/ether 00:0c:29:a5:4c:2b brd ff:ff:ff:ff:ff:ff
   inet 10.2.77.31/16 brd 10.2.255.255 scope global eth0
       valid_lft forever preferred_lft forever
   inet6 fe80::20c:29ff:fea5:4c2b/64 scope link
       valid_lft forever preferred_lft forever
\n Presione Enter para continuar...
```

```
==== Network Information Utility =====
1) Mostrar interfaces de red
2) Mostrar tabla de rutas
3) Mostrar conexiones activas
4) Mostrar estadísticas de red
5) Mostrar información de una interfaz
0) Salir
Seleccione una opción:
2
---- Routing Table ----
Kernel IP routing table
Destination    Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0        10.2.65.1      0.0.0.0         UG    0      0      0 eth0
10.2.0.0       0.0.0.0        255.255.0.0     U    0      0      0 eth0
127.0.0.0      0.0.0.0        255.0.0.0       U    0      0      0 lo
\n Presione Enter para continuar...
```



```

=====
Network Information Utility
=====
1) Mostrar interfaces de red
2) Mostrar tabla de rutas
3) Mostrar conexiones activas
4) Mostrar estadísticas de red
5) Mostrar información de una interfaz
0) Salir
Seleccione una opción:
3_

```

```

tcp6      0      0 :::5432                :::*                    LISTEN      1440/postgres
udp       0      0 0.0.0.0:161           0.0.0.0:*              1086/snmpd
udp       0      0 0.0.0.0:57626         0.0.0.0:*              1029/rpc.rquotad
udp       0 1536 10.2.77.31:58651       10.2.65.1:53           ESTABLISHED 1057/ntpd
udp       0      0 0.0.0.0:35119         0.0.0.0:*              -
udp       0      0 0.0.0.0:59842         0.0.0.0:*              1048/rpc.mountd
udp       0      0 10.2.77.31:43510      10.2.77.31:53          ESTABLISHED 1057/ntpd
udp       0      0 127.0.0.1:738         0.0.0.0:*              986/rpc.statd
udp       0      0 0.0.0.0:59129         0.0.0.0:*              1048/rpc.mountd
udp       0      0 10.2.77.31:53         0.0.0.0:*              1007/named
udp       0      0 10.2.77.31:53         0.0.0.0:*              1007/named
udp       0      0 127.0.0.1:53         0.0.0.0:*              1007/named
udp       0      0 127.0.0.1:53         0.0.0.0:*              1007/named
udp       0      0 0.0.0.0:54325         0.0.0.0:*              986/rpc.statd
udp       0      0 127.0.0.1:50264       127.0.0.1:53           ESTABLISHED 1057/ntpd
udp       0      0 0.0.0.0:45164         0.0.0.0:*              1048/rpc.mountd
udp       0      0 0.0.0.0:111          0.0.0.0:*              981/rpcbind
udp       0      0 10.2.77.31:123        0.0.0.0:*              1057/ntpd
udp       0      0 127.0.0.1:123        0.0.0.0:*              1057/ntpd
udp       0      0 0.0.0.0:123          0.0.0.0:*              1057/ntpd
udp6      0      0 :::49302              :::*                    1048/rpc.mountd
udp6      0      0 :::40172              :::*                    -
udp6      0      0 :::140259             :::*                    ESTABLISHED 1440/postgres
udp6      0      0 :::54656              :::*                    1029/rpc.rquotad
udp6      0      0 :::47770              :::*                    986/rpc.statd
udp6      0      0 :::58011              :::*                    1048/rpc.mountd
udp6      0      0 :::1:53               :::*                    1007/named
udp6      0      0 :::1:53               :::*                    1007/named
udp6      0      0 fe80::20c:29ff:fea5::53 :::*                    1007/named
udp6      0      0 fe80::20c:29ff:fea5::53 :::*                    1007/named
udp6      0      0 :::51268              :::*                    1048/rpc.mountd
udp6      0      0 :::111               :::*                    981/rpcbind
udp6      0      0 fe80::20c:29ff:fea5:123 :::*                    1057/ntpd
udp6      0      0 :::1:123              :::*                    1057/ntpd
udp6      0      0 :::123               :::*                    1057/ntpd
\n Presione Enter para continuar...

```

```

=====
Network Information Utility
=====
1) Mostrar interfaces de red
2) Mostrar tabla de rutas
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5) Mostrar información de una interfaz
0) Salir
Seleccione una opción:
4_

```

```
ICMP output histogram:
  destination unreachable: 346
IcmpMsg:
  InType3: 346
  OutType3: 346
Tcp:
  2 active connection openings
  0 passive connection openings
  2 failed connection attempts
  0 connection resets received
  0 connections established
  0 segments received
  2 segments sent out
  2 segments retransmitted
  0 bad segments received
  0 resets sent
Udp:
  1339 packets received
  0 packets to unknown port received
  0 packet receive errors
  1683 packets sent
  0 receive buffer errors
  0 send buffer errors
UdpLite:
TcpExt:
  0 packet headers predicted
  TCPTimeouts: 4
  TCPSpuriousRtxHostQueues: 2
  TCPSynRetrans: 2
  TcpTimeoutRehash: 4
IpExt:
  InOctets: 115905
  OutOctets: 137941
  InNoECTPkts: 1685
MPTcpExt:
\n Presione Enter para continuar...
```

```
=====
                        Network Information Utility
=====
1) Mostrar interfaces de red
2) Mostrar tabla de rutas
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4) Mostrar estadísticas de red
5) Mostrar información de una interfaz
0) Salir
Seleccione una opción:
5
```

```
Ingrese el nombre de la interfaz (ej. eth0, e1000g0):
eth0
---- Details for eth0 ----
Settings for eth0:
    Supported ports: [ TP ]
    Supported link modes:   10baseT/Half 10baseT/Full
                           100baseT/Half 100baseT/Full
                           1000baseT/Full
    Supported pause frame use: No
    Supports auto-negotiation: Yes
    Supported FEC modes: Not reported
    Advertised link modes:  10baseT/Half 10baseT/Full
                           100baseT/Half 100baseT/Full
                           1000baseT/Full
    Advertised pause frame use: No
    Advertised auto-negotiation: Yes
    Advertised FEC modes: Not reported
    Speed: 1000Mb/s
    Duplex: Full
    Auto-negotiation: on
    Port: Twisted Pair
    PHYAD: 0
    Transceiver: internal
    MDI-X: off (auto)
    Supports Wake-on: d
    Wake-on: d
    Current message level: 0x00000007 (7)
                           drv probe link

    Link detected: yes
\n Presione Enter para continuar...
```

## On Oracle Solaris:

```
root@solaris:~# sh network_info.sh
=====
Network Information Utility
=====
1) Mostrar interfaces de red
2) Mostrar tabla de rutas
3) Mostrar conexiones activas
4) Mostrar estadísticas de red
5) Mostrar información de una interfaz
0) Salir
Seleccione una opción:
1
---- Network Interfaces ----
lo0: flags=2001000849<UP,LOOPBACK,RUNNING,MULTICAST,IPv4,VIRTUAL> mtu 8232 index
  1
    inet 127.0.0.1 netmask ffffffff
net0: flags=100001000843<UP,BROADCAST,RUNNING,MULTICAST,IPv4,PHYSRUNNING> mtu 15
  00 index 2
    inet 10.2.77.35 netmask ffffff00 broadcast 10.2.255.255
    ether 0:c:29:76:d4:e4
lo0: flags=2002000849<UP,LOOPBACK,RUNNING,MULTICAST,IPv6,VIRTUAL> mtu 8252 index
  1
    inet6 ::1/128
net0: flags=120002000841<UP,RUNNING,MULTICAST,IPv6,PHYSRUNNING> mtu 1500 index 2
    inet6 fe80::20c:29ff:fe76:d4e4/10
    ether 0:c:29:76:d4:e4

Presione Enter para continuar...

```

```
=====
Network Information Utility
=====
1) Mostrar interfaces de red
2) Mostrar tabla de rutas
3) Mostrar conexiones activas
4) Mostrar estadísticas de red
5) Mostrar información de una interfaz
0) Salir
Seleccione una opción:
3

```

```
/system/volatile/hald/dbus-9tADby4dAw
ffffa1c013a3f550 stream-ord fffffa1c014286640 0000000000000000 /var/run/cups-sock
et
ffffa1c013a40a90 stream-ord 0000000000000000 0000000000000000 /system/volatile/h
ald/dbus-SROCBkYEDx
ffffa1c013a3f990 stream-ord 0000000000000000 fffffa1c01408a600
/system/volatile/hald/dbus-SROCBkYEDx
ffffa1c013a3fdd0 stream-ord fffffa1c01408a600 0000000000000000 /system/volatile/h
ald/dbus-SROCBkYEDx
ffffa1c013a40210 stream-ord fffffa1c01405cd80 0000000000000000 /system/volatile/i
netd.uds
ffffa1c00ae61008 stream-ord fffffa1c01403dd40 0000000000000000 /system/volatile/h
ald/dbus-9tADby4dAw
ffffa1c00ae61448 stream-ord 0000000000000000 0000000000000000 (socketpair)
(socketpair)
ffffa1c00ae61888 stream-ord 0000000000000000 0000000000000000 (socketpair)
(socketpair)
ffffa1c00ae61cc8 stream-ord fffffa1c0127f5180 0000000000000000 /var/run/dbus/syst
em_bus_socket
ffffa1c00ae62108 stream-ord fffffa1c01275ca00 0000000000000000 /system/volatile/r
ad/radsocket-unauth
ffffa1c00ae62548 stream-ord fffffa1c01275cc00 0000000000000000 /system/volatile/r
ad/radsocket
ffffa1c00ae62988 stream-ord fffffa1c01275ce00 0000000000000000 /system/volatile/r
ad/radsocket-unauth-http
ffffa1c00ae62dc8 stream-ord fffffa1c012736180 0000000000000000 /system/volatile/r
ad/radsocket-http
ffffa1c00ae63648 dgram fffffa1c00ae65b80 0000000000000000 /system/volatile/i
n.ndpd/in.ndpd_mib
ffffa1c00ae63a88 stream-ord fffffa1c00ae65d80 0000000000000000 /system/volatile/i
n.ndpd/in.ndpd_ipadm

Presione Enter para continuar...

```

```

=====
Network Information Utility
=====
1) Mostrar interfaces de red
2) Mostrar tabla de rutas
3) Mostrar conexiones activas
4) Mostrar estadísticas de red
5) Mostrar información de una interfaz
0) Salir
Seleccione una opción:
4

```

```

SCTP      sctpRtoAlgorithm    = vanj      sctpRtoMin      = 1000
          sctpRtoMax          = 60000    sctpRtoInitial  = 3000
          sctpMaxAssocs       = -1       sctpValCookieLife = 60000
          sctpMaxInitRetr     = 8         sctpCurrEstab   = 0
          sctpActiveEstab     = 0         sctpPassiveEstab = 0
          sctpAborted         = 0         sctpShutdowns   = 0
          sctpOutOfBlue       = 0         sctpChecksumError = 0
          sctpOutCtrlChunks   = 0         sctpOutOrderChunks = 0
          sctpOutUnorderChunks = 0        sctpRetransChunks = 0
          sctpOutAck          = 0         sctpOutAckDelayed = 0
          sctpOutWinUpdate    = 0         sctpOutFastRetrans = 0
          sctpOutWinProbe     = 0         sctpInCtrlChunks = 0
          sctpInOrderChunks   = 0         sctpInUnorderChunks = 0
          sctpInAck           = 0         sctpInDupAck     = 0
          sctpInAckUnsent     = 0         sctpFragUsrMsgs  = 0
          sctpReasmUsrMsgs    = 0         sctpOutSCTPPkts  = 0
          sctpInSCTPPkts      = 0         sctpInInvalidCookie = 0
          sctpTimRetrans      = 0         sctpTimRetransDrop = 0
          sctpTimHearBeatProbe = 0        sctpTimHearBeatDrop = 0
          sctpListenDrop      = 0         sctpInClosed     = 0

```

Presione Enter para continuar...

```

=====
Network Information Utility
=====
1) Mostrar interfaces de red
2) Mostrar tabla de rutas
3) Mostrar conexiones activas
4) Mostrar estadísticas de red
5) Mostrar información de una interfaz
0) Salir
Seleccione una opción:
5
Ingrese el nombre de la interfaz (ej. eth0, e1000g0):
net0
---- Details for net0 ----
net0: flags=100001000843<UP,BROADCAST,RUNNING,MULTICAST,IPv4,PHYSRUNNING> mtu 15
00 index 2
      inet 10.2.77.35 netmask ffff0000 broadcast 10.2.255.255
      ether 0:c:29:76:d4:e4

```

Presione Enter para continuar...

On Windows Server:

```
=====  
Network Information Utility  
=====
```

- 1) Show network interfaces
- 2) Show IP configuration
- 3) Show routing table
- 4) Show active connections
- 5) Show details of a specific interface
- 0) Exit

Select an option: 1  
---- Network Interfaces ----

| Name      | Status | LinkSpeed | MacAddress        |
|-----------|--------|-----------|-------------------|
| Ethernet0 | Up     | 1 Gbps    | 00-0C-29-67-1C-60 |

Press Enter to continue...: |

```
=====  
Network Information Utility  
=====
```

- 1) Show network interfaces
- 2) Show IP configuration
- 3) Show routing table
- 4) Show active connections
- 5) Show details of a specific interface
- 0) Exit

Select an option: 2  
---- IP Configuration ----

InterfaceAlias : Ethernet0  
InterfaceIndex : 13  
InterfaceDescription : Intel(R) 82574L Gigabit Network Connection  
NetProfile.Name : Red 2  
IPv4Address : 192.168.1.13  
IPv4DefaultGateway : 192.168.1.1  
DNSServer : 192.168.1.1

Press Enter to continue...: |

```
=====  
Network Information Utility  
=====
```

- 1) Show network interfaces
- 2) Show IP configuration
- 3) Show routing table
- 4) Show active connections
- 5) Show details of a specific interface
- 0) Exit

Select an option: 3

```
---- Routing Table ----

ifIndex DestinationPrefix NextHop RouteMetric ifMetric PolicyStore
-----
13 255.255.255.255/32 0.0.0.0 256 25 ActiveStore
1 255.255.255.255/32 0.0.0.0 256 75 ActiveStore
13 224.0.0.0/4 0.0.0.0 256 25 ActiveStore
1 224.0.0.0/4 0.0.0.0 256 75 ActiveStore
13 192.168.1.255/32 0.0.0.0 256 25 ActiveStore
13 192.168.1.13/32 0.0.0.0 256 25 ActiveStore
13 192.168.1.0/24 0.0.0.0 256 25 ActiveStore
1 127.255.255.255/32 0.0.0.0 256 75 ActiveStore
1 127.0.0.1/32 0.0.0.0 256 75 ActiveStore
1 127.0.0.0/8 0.0.0.0 256 75 ActiveStore
13 0.0.0.0/0 192.168.1.1 0 25 ActiveStore
1 ff00::/8 :: 256 75 ActiveStore
1 ::1/128 :: 256 75 ActiveStore

Press Enter to continue...:
```

```
Administrator: Windows Pow X + v
=====
Network Information Utility
=====
1) Show network interfaces
2) Show IP configuration
3) Show routing table
4) Show active connections
5) Show details of a specific interface
0) Exit
Select an option: 4
---- Active TCP/UDP Connections ----

LocalAddress LocalPort RemoteAddress RemotePort State AppliedSetting OwningProcess
-----
:: 49670 :: 0 Listen 888
:: 49669 :: 0 Listen 3308
:: 49668 :: 0 Listen 1548
:: 49667 :: 0 Listen 2352
:: 49666 :: 0 Listen 1864
:: 49665 :: 0 Listen 768
:: 49664 :: 0 Listen 912
:: 47001 :: 0 Listen 4
:: 5985 :: 0 Listen 4
::1 1434 :: 0 Listen 2072
:: 1433 :: 0 Listen 2072
:: 445 :: 0 Listen 4
:: 135 :: 0 Listen 600
:: 80 :: 0 Listen 4
::1 53 :: 0 Listen 3308
0.0.0.0 62423 0.0.0.0 0 Bound 2928
0.0.0.0 59207 0.0.0.0 0 Bound 3364
0.0.0.0 58418 0.0.0.0 0 Bound 2928
192.168.1.13 62423 104.208.203.89 443 Established Internet 2928
192.168.1.13 59207 52.123.128.14 443 Established Datacenter 3364
192.168.1.13 58418 104.208.203.89 443 Established Internet 2928
```

```
LocalAddress LocalPort
-----
:: 63864
:: 63584
:: 60366
:: 55756
::1 51046
:: 50719
:: 123
::1 53
0.0.0.0 63863
0.0.0.0 53548
0.0.0.0 53547
0.0.0.0 53546
0.0.0.0 53545
0.0.0.0 53544
```

```
Administrator: Windows Pow  X  +  v  -  □  X

=====
Network Information Utility
=====
1) Show network interfaces
2) Show IP configuration
3) Show routing table
4) Show active connections
5) Show details of a specific interface
0) Exit
Select an option: 5
Enter interface name: Ethernet0
---- Details for Ethernet0 ----

Name                : Ethernet0
InterfaceDescription : Intel(R) 82574L Gigabit Network Connection
InterfaceIndex       : 13
MacAddress           : 00-0C-29-67-1C-60
MediaType            : 802.3
PhysicalMediaType    : 802.3
InterfaceOperationalStatus : Up
AdminStatus          : Up
LinkSpeed(Gbps)      : 1
MediaConnectionState : Connected
ConnectorPresent     : True
DriverInformation     : Driver Date 2015-08-03 Version 12.19.1.32 NDIS 6.50
```

b. Additionally, create a Shell program that indicates (yes or no) whether a port is open and which service is running on that port.

On Linux Slackware:

```
root@aysrSlack:~# sh check_port.sh
Ingrese el número de puerto a verificar:
80
Verificando puerto 80 en Linux...
El puerto 80 está ABIERTO.
Servicio asociado: http
root@aysrSlack:~# _

root@aysrSlack:~# sh check_port.sh
Ingrese el número de puerto a verificar:
161
Verificando puerto 161 en Linux...
El puerto 161 está ABIERTO.
Servicio asociado: snmp
root@aysrSlack:~#

root@aysrSlack:~# sh check_port.sh
Ingrese el número de puerto a verificar:
22
Verificando puerto 22 en Linux...
El puerto 22 está ABIERTO.
Servicio asociado: ssh
root@aysrSlack:~#

root@aysrSlack:~# sh check_port.sh
Ingrese el número de puerto a verificar:
8080
Verificando puerto 8080 en Linux...
El puerto 8080 está CERRADO o no se encontró en uso.
root@aysrSlack:~#
```

On Oracle Solaris:

```
root@solaris:~# sh check_port.sh
Ingrese el n mero de puerto a verificar:
80
Verificando puerto 80 en SunOS...
El puerto 80 est  ABIERTO.
Servicio asociado: http
root@solaris:~#

root@solaris:~# sh check_port.sh
Ingrese el n mero de puerto a verificar:
161
Verificando puerto 161 en SunOS...
El puerto 161 est  ABIERTO.
Servicio asociado: snmpd
root@solaris:~#

root@solaris:~# sh check_port.sh
Ingrese el n mero de puerto a verificar:
123
Verificando puerto 123 en SunOS...
El puerto 123 est  CERRADO o no se encontr  en uso.
root@solaris:~#
```

On Windows Server:

```
Administrador: Windows Pow  x + v
Enter the port number to check: 80
Checking port 80...
Port 80 is OPEN.
Service associated: http      80/tcp    www www-http      #World Wide Web
PS C:\Users\Administrador\downloads>

Administrador: Windows Pow  x + v
Enter the port number to check: 8080
Checking port 8080...
Port 8080 is CLOSED or not in LISTEN state.
PS C:\Users\Administrador\downloads>
```

## 2. Network Management

On the other hand, part of an organization's technological platform is the monitoring server. Through it, administrators can check the status of the network equipment they manage. These platform management tools enable remote monitoring of disk space, CPU usage, network performance, memory usage, installed software, among other aspects of the network devices.

Each student must install a server to monitor the network (Groups of 2 students: one student should use Slackware and the other Solaris. Groups of 3 students: additionally, the third student will use Windows) and document their work. They must enable monitoring of all their virtual machines and display it in real-time.

Show your professor the configurations, ports, and monitoring software installations.

Hint: Look for SNMP protocol implementations.



On Oracle Solaris:

First we install the SNMP service

```
root@solaris:~# pkg install net-snmp
No hay actualizaciones para esta imagen.
root@solaris:~#
```

Create the /etc/snmp directory

```
root@solaris:~# mkdir -p /etc/snmp
root@solaris:~#
```

We created the file /etc/snmp/snmpd.conf and added the following

```
GNU nano 2.9.3 /etc/snmp/snmpd.conf Modified
# /etc/snmp/snmpd.conf - Basic SNMP configuration for Solaris

# Solo lectura con comunidad 'public'
rocommunity public default

# Información del sistema
syslocation "Solaris VM"
syscontact "admin@empresa.local"

# Monitoreo básico de recursos
disk / 10%
load 12 10 5
|
```

We enabled the SNMP service and verified

```
root@solaris:~# svcadm enable svc:/application/management/net-snmp:default
root@solaris:~# svcs -a | grep snmp
disabled      3:58:33 svc:/system/fm/snmp-notify:default
online        4:22:42 svc:/application/management/net-snmp:default
root@solaris:~#
```

We verified that it is listening on port 161

```
root@solaris:~# netstat -an | grep 161
*.161               Idle                57344              0              57344              0
root@solaris:~#
```

We tested locally that SNMP was working

```
root@solaris:~# snmpwalk -v2c -c public localhost
SNMPv2-MIB::sysDescr.0 = STRING: SunOS solaris 5.11 11.4.0.15.0 i86pc
SNMPv2-MIB::sysObjectID.0 = OID: NET-SNMP-MIB::netSnmpAgentOIDs.3
DISMAN-EVENT-MIB::sysUpTimeInstance = Timeticks: (36094) 0:06:00.94
SNMPv2-MIB::sysContact.0 = STRING: \"System administrator\"
SNMPv2-MIB::sysName.0 = STRING: solaris
SNMPv2-MIB::sysLocation.0 = STRING: \"System administrators office\"
SNMPv2-MIB::sysServices.0 = INTEGER: 72
SNMPv2-MIB::sysORLastChange.0 = Timeticks: (401) 0:00:04.01
SNMPv2-MIB::sysORID.1 = OID: SNMP-MPD-MIB::snmpMPDCompliance
SNMPv2-MIB::sysORID.2 = OID: SNMP-USER-BASED-SM-MIB::usmMIBCompliance
SNMPv2-MIB::sysORID.3 = OID: SNMP-FRAMEWORK-MIB::snmpFrameworkMIBCompliance
SNMPv2-MIB::sysORID.4 = OID: SNMPv2-MIB::snmpMIB
```

We allow queries to the server from within machines on the same network

```
GNU nano 2.9.3 /etc/snmp/snmpd.conf Modified
# /etc/snmp/snmpd.conf - Basic SNMP configuration for Solaris

# Solo lectura con comunidad 'public'
rocommunity public 10.2.0.0/16

# Información del sistema
syslocation "Solaris VM"
syscontact "admin@empresa.local"

# Monitoreo básico de recursos
disk / 10%
load 12 10 5
```

Then restart the service

```
root@solaris:~# svcadm restart svc:/application/management/net-snmp:default
root@solaris:~# |
```

We installed the monitoring system on Solaris

```
root@solaris:/var/www# wget https://files.cacti.net/cacti/linux/cacti-1.2.25.tar.gz
--2025-11-12 06:07:36-- https://files.cacti.net/cacti/linux/cacti-1.2.25.tar.gz
Resolving files.cacti.net (files.cacti.net)... 23.166.80.12, 64:ff9b::17a6:500c
Connecting to files.cacti.net (files.cacti.net)|23.166.80.12|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 43303397 (41M) [application/octet-stream]
Saving to: 'cacti-1.2.25.tar.gz'

cacti-1.2.25.tar.gz      100%[=====] 41.30M  331KB/s  in 1m 43s

2025-11-12 06:09:20 (409 KB/s) - 'cacti-1.2.25.tar.gz' saved [43303397/43303397]

root@solaris:/var/www#
```

We unzip and send the contents of the folder to the cacti folder

```
root@solaris:/var/www# tar xzf cacti-1.2.25.tar.gz
root@solaris:/var/www# mv cacti-1.2.25 cacti
root@solaris:/var/www# |
```

We run the following command to ensure that Apache has access to them and change the permissions

```
root@solaris:/var/www/cacti# chown -R webservd:webservd /var/www/cacti
root@solaris:/var/www/cacti# chmod -R 755 /var/www/cacti
root@solaris:/var/www/cacti# |
```

Edit /var/www/cacti/include/config.php to configure mysql

```
GNU nano 2.9.3 /var/www/cacti/include/config.php Modified
<?php
/*
+-----+
| Copyright (C) 2004-2023 The Cacti Group
|
| This program is free software; you can redistribute it and/or
| modify it under the terms of the GNU General Public License
| as published by the Free Software Foundation; either version 2
| of the License, or (at your option) any later version.
|
| This program is distributed in the hope that it will be useful,
| but WITHOUT ANY WARRANTY; without even the implied warranty of
| MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the
| GNU General Public License for more details.
+-----+
| Cacti: The Complete RRDtool-based Graphing Solution
+-----+
| This code is designed, written, and maintained by the Cacti Group. See
| about.php and/or the AUTHORS file for specific developer information.
+-----+
| http://www.cacti.net/
+-----+
*/

/**
 * Make sure these values reflect your actual database/host/user/password
 */

$database_type      = 'mysql';
$database_default   = 'cacti';
$database_hostname   = 'localhost';
$database_username   = 'cactiuser';
$database_password   = 'admin123';
$database_port       = '3306';
$database_retries    = 5;
$database_ssl        = false;
$database_ssl_key    = '';
$database_ssl_cert    = '';
$database_ssl_ca     = '';
$database_persist    = false;
```

Now restart apache

```
root@solaris:/var/www/cacti# svcadm restart apache24
root@solaris:/var/www/cacti# svcadm enable svc:/network/http:apache24
root@solaris:/var/www/cacti# svcadm disable svc:/network/http:apache24
root@solaris:/var/www/cacti# svcadm enable svc:/network/http:apache24
root@solaris:/var/www/cacti# svcs -a | grep -i http
online      8:01:56 svc:/network/http:apache24
root@solaris:/var/www/cacti#
```

Create the file /etc/apache2/2.4/conf.d/cacti.conf and add this

```
GNU nano 2.9.3 /etc/apache2/2.4/conf.d/cacti.conf Modified
Alias /cacti /var/www/cacti

<Directory /var/www/cacti>
    Options FollowSymLinks
    AllowOverride All
    Require all granted
</Directory>
```

```

root@solaris:~# ls /etc/apache2/2.4/conf.d | grep php
root@solaris:~# echo 'LoadModule php7_module libexec/apache2/mod_php7.1.so' >> /etc/apache2/2.4/httpd.conf
root@solaris:~# svcadm restart apache24
root@solaris:~# svcs apache24
STATE          STIME          FMRI
online         9:05:26        svc:/network/http:apache24
root@solaris:~#

```

Now we look for which version of Solaris is available for MySQL and install it.

```

root@solaris:~# pkg search mysql-57
INDEX      ACTION VALUE          PACKAGE
pkg.fmri   set      solaris/database/mysql-57 pkg:/database/mysql-57@5.7.22-11.4.0.0.1.14.0
root@solaris:~# pkg install mysql-57
Packages to install: 3
Mediators to change: 1
Services to change: 1
Create boot environment: No
Create backup boot environment: No

DOWNLOAD                                PKGS      FILES    XFER (MB)   SPEED
Completed                              3/3       314/314   231.0/231.0 2.9M/s

PHASE                                ITEMS
Installing new actions                379/379
Updating package state database        Done
Updating package cache                 0/0
Updating image state                   Done
Creating fast lookup database          Done
Updating package cache                 1/1
root@solaris:~# |

```

Now we start the service and verify

```

root@solaris:~# svcadm enable mysql
root@solaris:~# svcs -a | grep mysql
online         6:44:29        svc:/application/database/mysql:version_57
root@solaris:~# |

```

Now, to configure MySQL, we disable the service and verify.

```

root@solaris:~# svcadm disable mysql
root@solaris:~# svcs -a | grep mysql
disabled       6:47:57        svc:/application/database/mysql:version_57
root@solaris:~# |

```

Now we start MySQL in safe mode and log in without a password.

```

root@solaris:~# /usr/mysql/5.7/bin/mysqld_safe --skip-grant-tables &
[1] 1430
root@solaris:~# Logging to '/var/mysql/5.7/solaris.err'.
2025-11-21T11:50:12.258933Z mysqld_safe Starting mysqld daemon with databases from /var/mysql/5.7
^C
root@solaris:~# mysql -u root
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 2
Server version: 5.7.22 MySQL Community Server (GPL)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> |

```

## We initialize MySQL

```
root@solaris:~# /usr/mysql/5.7/bin/mysqld --initialize-insecure --user=mysql --datadir=/var/mysql/5.7/data
2025-11-21T12:18:16.516890Z 0 [Warning] TIMESTAMP with implicit DEFAULT value is deprecated. Please use --explicit_defaults_for_timestamp server option (see documentation for more details).
2025-11-21T12:18:16.802743Z 0 [Warning] InnoDB: New log files created, LSN=45790
2025-11-21T12:18:16.834120Z 0 [Warning] InnoDB: Creating foreign key constraint system tables.
2025-11-21T12:18:16.892163Z 0 [Warning] No existing UUID has been found, so we assume that this is the first time that this server has been started. Generating a new UUID: 28d555c4-c6d4-11f0-a43a-81fb6b226bb3.
2025-11-21T12:18:16.892992Z 0 [Warning] Gtid table is not ready to be used. Table 'mysql.gtid_executed' cannot be opened.
2025-11-21T12:18:16.893841Z 1 [Warning] root@localhost is created with an empty password ! Please consider switching off the --initialize-insecure option.
```

## Now we enable the service and verify

```
root@solaris:~# svcadm enable mysql:version_57
root@solaris:~# svcs -xv mysql:version_57
svc:/application/database/mysql:version_57 (MySQL Database Management System)
  State: online since Fri Nov 21 07:14:36 2025
  See: http://dev.mysql.com/doc
  See: man -M /usr/share/man -s 1 MySQL 5.7
  See: /var/svc/log/application-database-mysql:version_57.log
Impact: None.
root@solaris:~#
```

## We start MySQL

```
root@solaris:~# mysql -u root
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 3
Server version: 5.7.22 MySQL Community Server (GPL)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

## Now we create the Cacti database and a dedicated user for this database

```
mysql> CREATE DATABASE cacti;
Query OK, 1 row affected (0.00 sec)

mysql> CREATE USER 'cacti_user'@'localhost' IDENTIFIED BY 'admin123';
Query OK, 0 rows affected (0.00 sec)

mysql> GRANT ALL PRIVILEGES ON cacti.* TO 'cacti_user'@'localhost';
Query OK, 0 rows affected (0.00 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.00 sec)

mysql>
```

## We look for the cacti.sql file and import it

```
root@solaris:~# find / -name "cacti.sql" 2>/dev/null
/var/www/cacti/cacti.sql
root@solaris:~# mysql -u root cacti < /var/www/cacti/cacti.sql
```

Then we went into MySQL and checked the import.

```
root@solaris:/# mysql -u root
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 5
Server version: 5.7.22 MySQL Community Server (GPL)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE cacti;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

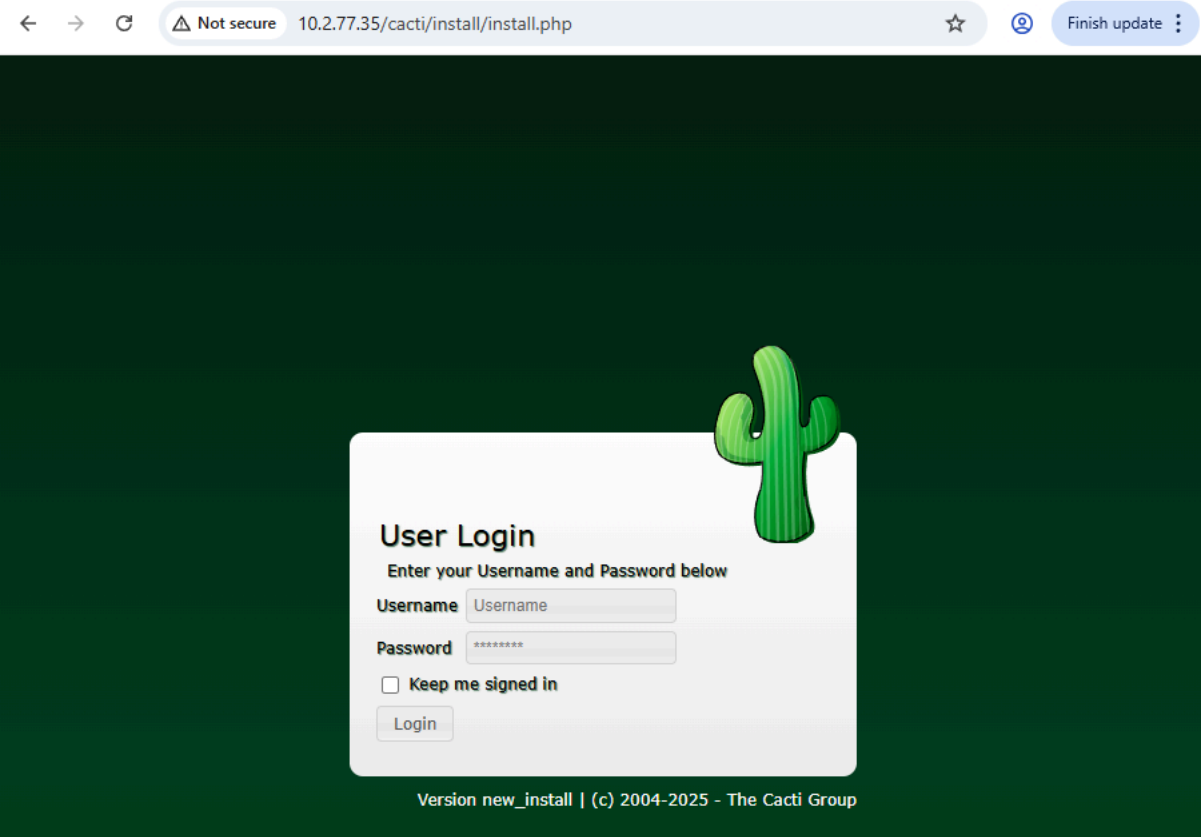
Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_cacti |
+-----+
| aggregate_graph_templates |
| aggregate_graph_templates_graph |
| aggregate_graph_templates_item |
| aggregate_graphs |
| aggregate_graphs_graph_item |
| aggregate_graphs_items |
| automation_devices |
| automation_graph_rule_items |
| automation_graph_rules |
| automation_ips |
| automation_match_rule_items |
| automation_networks |
| automation_processes |
| automation_snmp |
| automation_snmp_items |
| automation_templates |
| automation_tree_rule_items |
| automation_tree_rules |
+-----+
```

Edit /etc/php/7.1/conf.d/extensions.ini and we uncomment the two lines

```
GNU nano 2.9.3 /etc/php/7.1/conf.d/extensions.ini Modified
;
; Copyright (c) 2015, 2017, Oracle and/or its affiliates. All rights reserved.
;
; To enable PHP extensions, uncomment the desired line(s) below, or add similar
; lines to another active PHP configuration (.ini) file.

;extension=bcmath.so
;extension=bz2.so
;extension=calendar.so
;extension=curl.so
;extension=dba.so
;extension=enchant.so
;extension=exif.so
;extension=ftp.so
;extension=gd.so
;extension=gettext.so
;extension=gmp.so
;extension=intl.so
;extension=ldap.so
;extension=mbstring.so
extension=mysqli.so
;extension=opcache.so
;extension=openssl.so
;extension=pcntl.so
extension=pdo_mysql.so
;extension=readline.so
;extension=shmop.so
;extension=snmp.so
;extension=soap.so
;extension=sockets.so
;extension=sysvmsg.so
;extension=sysvsem.so
;extension=sysvshm.so
;extension=tidy.so
```

Login as admin and password admin



← → ↻ ⚠ Not secure 10.2.77.35/cacti/install/install.php ☆ ⓘ Finish update ⋮

### User Login

Enter your Username and Password below

Username

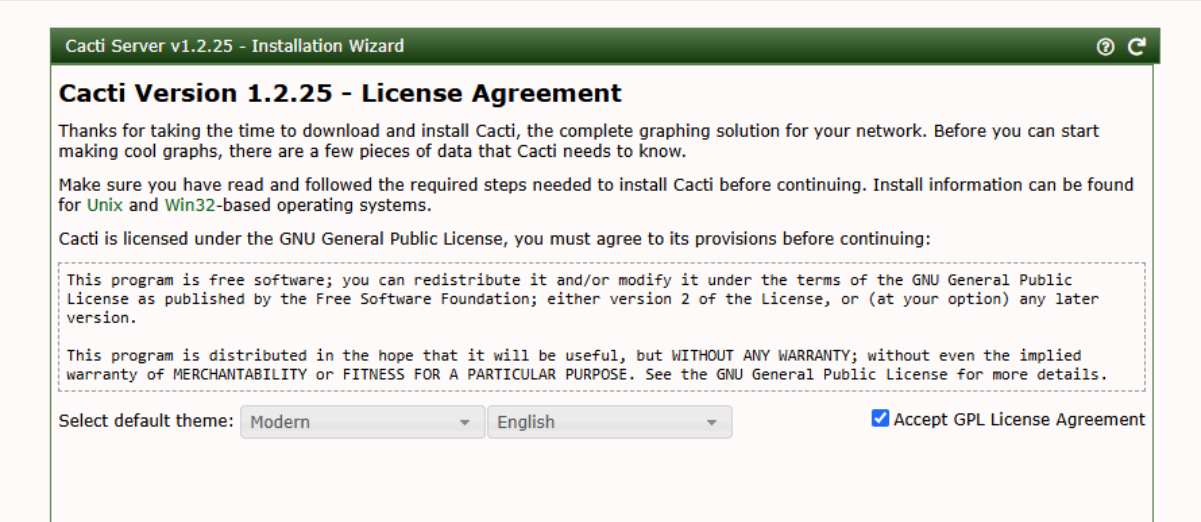
Password

☐ Keep me signed in

Login

Version new\_install | (c) 2004-2025 - The Cacti Group

Change the password and accept GPL license agreement



Cacti Server v1.2.25 - Installation Wizard ⓘ ⌂

### Cacti Version 1.2.25 - License Agreement

Thanks for taking the time to download and install Cacti, the complete graphing solution for your network. Before you can start making cool graphs, there are a few pieces of data that Cacti needs to know.

Make sure you have read and followed the required steps needed to install Cacti before continuing. Install information can be found for [Unix](#) and [Win32](#)-based operating systems.

Cacti is licensed under the GNU General Public License, you must agree to its provisions before continuing:

This program is free software; you can redistribute it and/or modify it under the terms of the GNU General Public License as published by the Free Software Foundation; either version 2 of the License, or (at your option) any later version.

This program is distributed in the hope that it will be useful, but WITHOUT ANY WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the GNU General Public License for more details.

Select default theme: Modern ▾ English ▾ ☒ Accept GPL License Agreement

Edit /etc/php/7.1/php.ini as follows

```
[Date]
; Defines the default timezone used by the date functions
; http://php.net/date.timezone
date.timezone = America/Bogota
```

We provide access to the timezone in MySQL

```
root@solaris:/# mysql -u root
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 29
Server version: 5.7.22 MySQL Community Server (GPL)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> GRANT SELECT ON mysql.time_zone_name TO 'cacti_user'@'localhost';
Query OK, 0 rows affected (0.00 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.00 sec)

mysql> exit
Bye
root@solaris:/# |
```

Edit /etc/php/7.1/php.ini

```
; Maximum amount of memory a script may consume (128MB)
; http://php.net/memory-limit
memory_limit = 512M
```

On Linux Slackware

We installed SNMP using DCRON

```
root@aysrSlack:/mnt/cdrom/slackware64/n# installpkg net-snmp-5.9.1-x86_64-4.txz
Verifying package net-snmp-5.9.1-x86_64-4.txz.
Installing package net-snmp-5.9.1-x86_64-4.txz [REC]:
PACKAGE DESCRIPTION:
# net-snmp (Simple Network Management Protocol tools)
#
# Various tools relating to the Simple Network Management Protocol:
#
# An extensible agent
# An SNMP library
# Tools to request or set information from SNMP agents
# Tools to generate and handle SNMP traps
# A version of the UNIX 'netstat' command using SNMP
# A graphical Perl/Tk/SNMP based mib browser
#
Executing install script for net-snmp-5.9.1-x86_64-4.txz.
Package net-snmp-5.9.1-x86_64-4.txz installed.
root@aysrSlack:/mnt/cdrom/slackware64/n#
```

We verified the installation and version

```
root@aysrSlack:/mnt/cdrom/slackware64/n# which snmpd
/usr/sbin/snmpd
```

```
root@aysrSlack:/mnt/cdrom/slackware64/n# snmpd --version

NET-SNMP version: 5.9.1
Web: http://www.net-snmp.org/
Email: net-snmp-coders@lists.sourceforge.net

root@aysrSlack:/mnt/cdrom/slackware64/n#
```



We edit /etc/snmp/snmpd.conf as follows

```
rocommunity public 10.2.0.0/16

#####
# System contact information
#

# It is also possible to set the sysContact and sysLocation system
# variables through the snmpd.conf file:
syslocation "Slackware Monitoring Server"
syscontact "Admin <admin@aysr.local>"
syslocation Unknown (edit /etc/snmp/snmpd.conf)
syscontact Root <root@localhost> (configure /etc/snmp/snmp.local.conf)

view systemview included .1.3.6.1.2.1.1
view systemview included .1.3.6.1.2.1.25.1.1
access notConfigGroup "" any noauth exact systemview none none
```

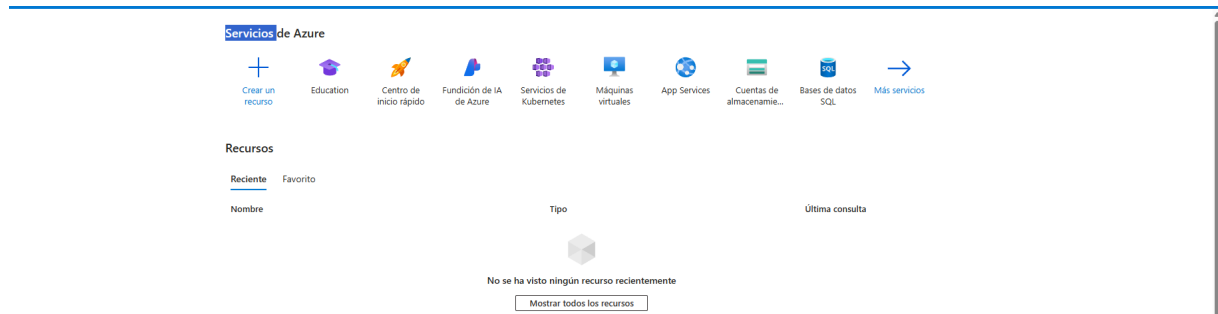
We verified the port status and tested the functionality of the SNMP server.

```
root@aysrSlack:/mnt/cdrom/slackware64/n# /usr/sbin/snmpd -c /etc/snmp/snmpd.conf
root@aysrSlack:/mnt/cdrom/slackware64/n# netstat -tulnp | grep 161
udp        0      0 0.0.0.0:0:161        0.0.0.0:*                1087/snmpd
udp6       0      0 ::::161              :::*                      2575/snmpd
root@aysrSlack:/mnt/cdrom/slackware64/n# snmpwalk -v2c -c public localhost
SNMPv2-MIB::sysDescr.0 = STRING: Linux aysrSlack.local 5.15.19 #1 SMP PREEMPT Wed Feb 2 01:50:51 CST 2022 x86_64
SNMPv2-MIB::sysObjectID.0 = OID: NET-SNMP-MIB::netSnmpAgentOIDs.10
DISMAN-EVENT-MIB::sysUpTimeInstance = Timeticks: (713410) 1:58:54.10
SNMPv2-MIB::sysContact.0 = STRING: Root <root@localhost> (configure /etc/snmp/snmp.local.conf)
SNMPv2-MIB::sysName.0 = STRING: aysrSlack.local
SNMPv2-MIB::sysLocation.0 = STRING: Unknown (edit /etc/snmp/snmpd.conf)
SNMPv2-MIB::sysORLastChange.0 = Timeticks: (19) 0:00:00.19
SNMPv2-MIB::sysORID.1 = OID: SNMP-FRAMEWORK-MIB::snmpFrameworkMIBCompliance
SNMPv2-MIB::sysORID.2 = OID: SNMP-MPD-MIB::snmpMPDCompliance
SNMPv2-MIB::sysORID.3 = OID: SNMP-USER-BASED-SM-MIB::usmMIBCompliance
SNMPv2-MIB::sysORID.4 = OID: SNMPv2-MIB::snmpMIB
SNMPv2-MIB::sysORID.5 = OID: SNMP-VIEW-BASED-ACM-MIB::vacmBasicGroup
SNMPv2-MIB::sysORID.6 = OID: TCP-MIB::tcpMIB
SNMPv2-MIB::sysORID.7 = OID: UDP-MIB::udpMIB
SNMPv2-MIB::sysORID.8 = OID: IP-MIB::ip
SNMPv2-MIB::sysORID.9 = OID: SNMP-NOTIFICATION-MIB::snmpNotifyFullCompliance
SNMPv2-MIB::sysORID.10 = OID: NOTIFICATION-LOG-MIB::notificationLogMIB
SNMPv2-MIB::sysORDescr.1 = STRING: The SNMP Management Architecture MIB.
SNMPv2-MIB::sysORDescr.2 = STRING: The MIB for Message Processing and Dispatching.
SNMPv2-MIB::sysORDescr.3 = STRING: The management information definitions for the SNMP User-based Security Model.
SNMPv2-MIB::sysORDescr.4 = STRING: The MIB module for SNMPv2 entities
SNMPv2-MIB::sysORDescr.5 = STRING: View-based Access Control Model for SNMP.
SNMPv2-MIB::sysORDescr.6 = STRING: The MIB module for managing TCP implementations
SNMPv2-MIB::sysORDescr.7 = STRING: The MIB module for managing UDP implementations
SNMPv2-MIB::sysORDescr.8 = STRING: The MIB module for managing IP and ICMP implementations
SNMPv2-MIB::sysORDescr.9 = STRING: The MIB modules for managing SNMP Notification, plus filtering.
SNMPv2-MIB::sysORDescr.10 = STRING: The MIB module for logging SNMP Notifications.
SNMPv2-MIB::sysORUpTime.1 = Timeticks: (11) 0:00:00.11
SNMPv2-MIB::sysORUpTime.2 = Timeticks: (11) 0:00:00.11
SNMPv2-MIB::sysORUpTime.3 = Timeticks: (11) 0:00:00.11
SNMPv2-MIB::sysORUpTime.4 = Timeticks: (11) 0:00:00.11
```

### 3. Network Administration - Azure

1. Log in to the Azure portal.

We log in with the institutional mail



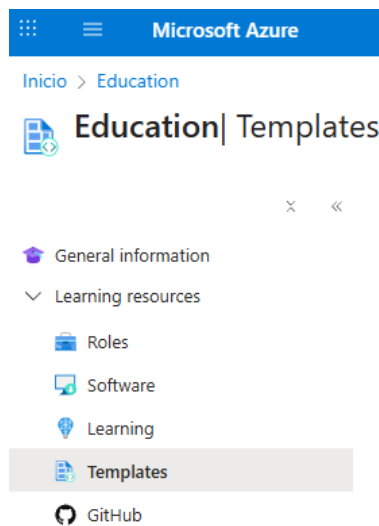
2. Navigate to the "Education" section.

We go to the education section.



3. Go to the "Templates" section.

Then we go to the templates section



#### 4. Select the template named "Web App Deployment from GitHub."

|                                |                   |              |           |      |
|--------------------------------|-------------------|--------------|-----------|------|
| Web.App.Deployment.from.GitHub | Azure App Service | Intermediate | Developer | FREE |
|--------------------------------|-------------------|--------------|-----------|------|

We select the required option.

#### 5. Deploy the selected template.

#### 6. Choose an existing resource or create a new one, then click "Review + create" and subsequently "Create."

We deployed and created the template

[Básico](#) [Revisar y crear](#)

**Plantilla**  
 [Plantilla personalizada](#)   
3 recursos

[Editar plantilla](#) [Editar paráme...](#) [Visualizar](#)

**Detalles del proyecto**  
Seleccione la suscripción para administrar recursos implementados y los costes. Use los grupos de recursos como carpetas para organizar y administrar todos los recursos.

Suscripción \* ⓘ

Suscripción de Azure 1 ▼

Grupo de recursos \* ⓘ

(Nuevo) Lab7JulianSantiago ▼

[Crear nuevo](#)

**Detalles de la instancia**

Región \* ⓘ

West US 2 ▼

Site Name ⓘ

labJulian

Location ⓘ

[resourceGroup().location]

Skus ⓘ

F1 ▼

Worker Size ⓘ

0 ▼

Repo URL ⓘ

<https://github.com/Azure-Samples/app-service-web-html-get-started....> ✓

Branch ⓘ

master ✓

< Anterior

Siguiente

Revisar y crear

#### 7. Navigate to the created resource and access the "Web App". Explore the website by clicking the "Default domain" at the "Overview" section.

We click in "Dominio predeterminado"

 **Dominios**

Dominio predeterminado

labjulianamaya.azurewebsites.net

Dominio personalizado

[Agregar dominio personalizado](#)

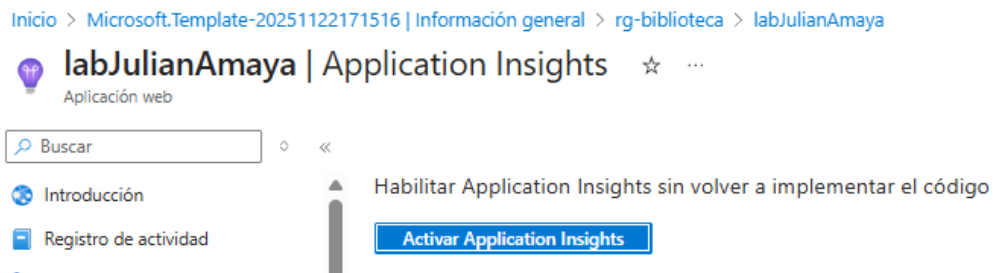
 **Application Insights**

Nombre

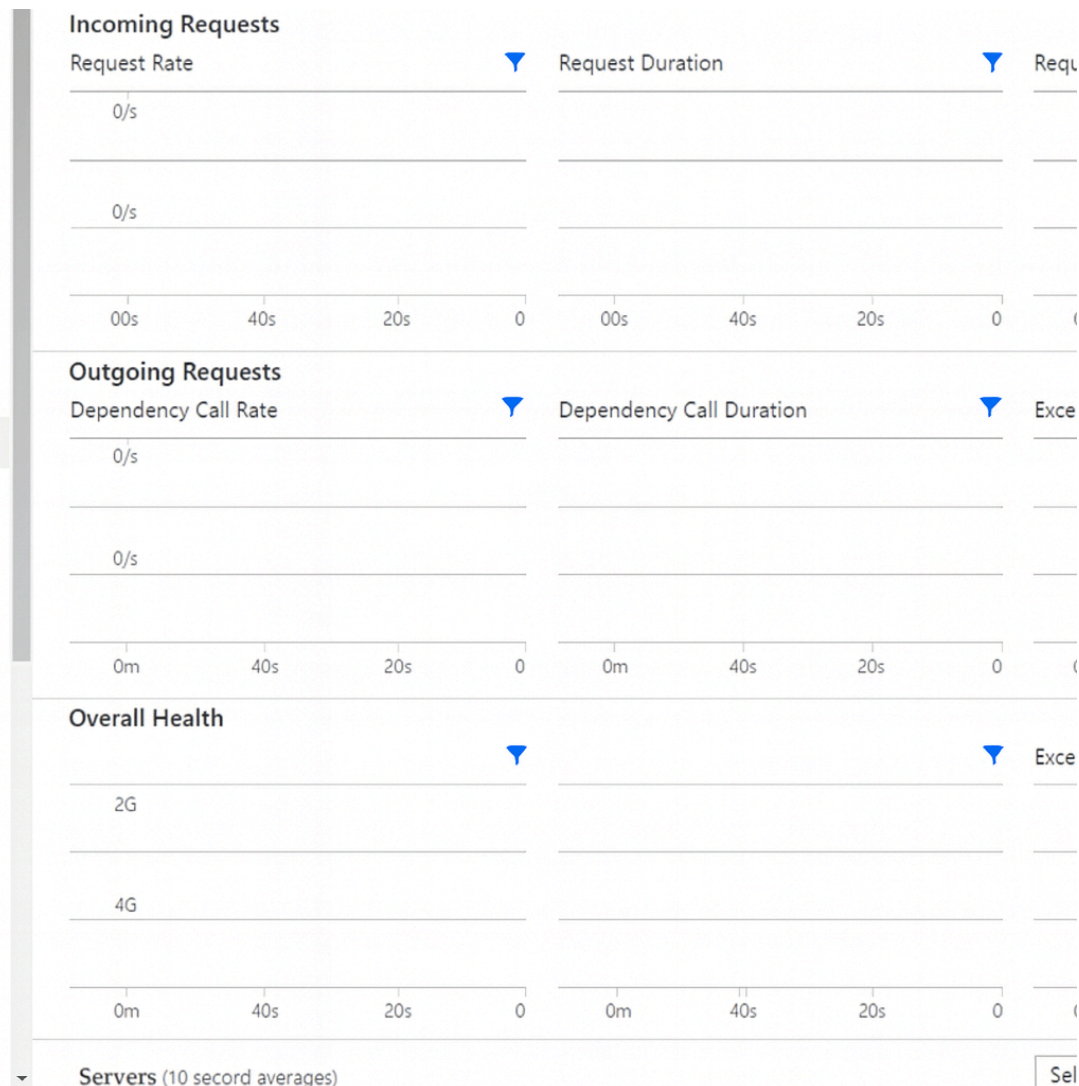
[Habilitar Application Insights](#)

8. Now, go to the "Monitoring" section, then navigate to "Application Insights" and enable the service if it isn't already active. Once there, click on "View Application Insights data."

First we Activate the service

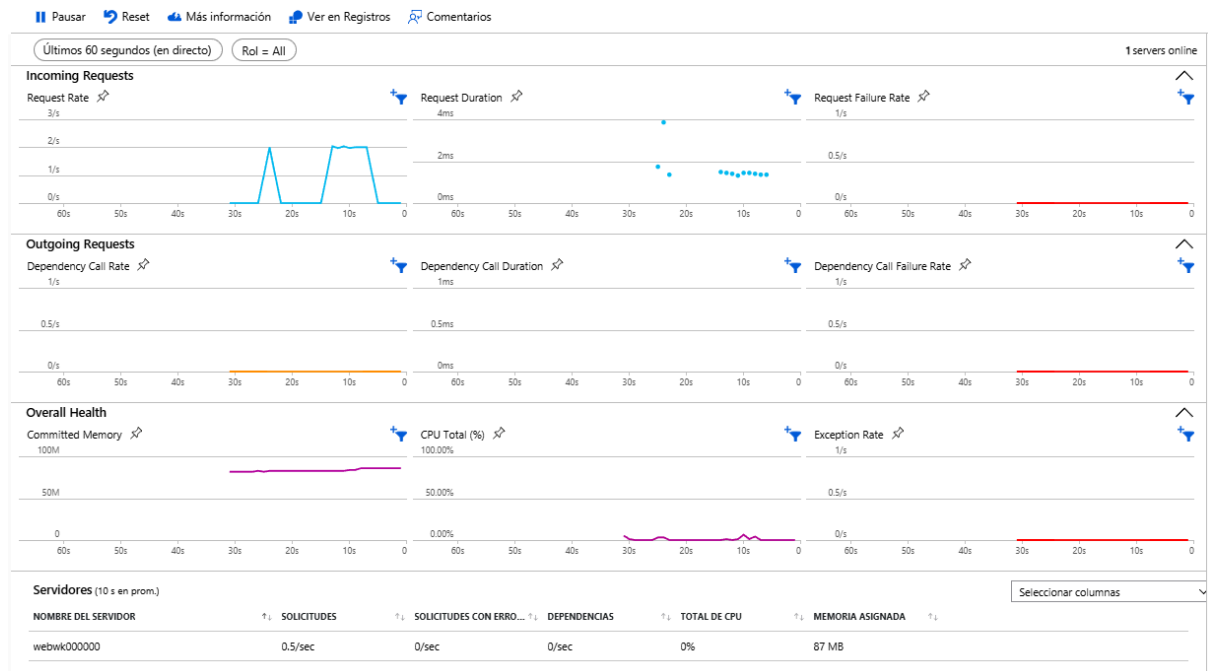


9. Explore the "Overview" and "Live Metric" tabs in the "Investigate" section. Because it's a new webpage it shows everything in white, and we don't have any activities yet.



– What do you observe when you refresh the website repeatedly?

We see some new activity that represents our server working and enabled.



– What does each of the items displayed there represent?

The Application Insights Live Metrics panel provides a real-time view of the web application's performance and health, grouping the telemetry into several essential categories for application performance monitoring.

The incoming requests section tracks traffic arriving from users, this area details the request rate the server is receiving per second, the duration it takes the server to respond to each request, and the failure rate in communication.

The Outgoing Requests measure the performance of calls the application makes to external services, these services may include databases, caches, or third-party APIs. This section reports the call rate and duration of these external interactions, identifying potential bottlenecks outside the application itself.

Overall Health section monitors critical resources of the underlying server, it displays the total CPU usage and the memory allocated to the process, in addition to the code's exception rate, offering a complete, real-time diagnosis of the application's activity and stability.

– What other functionalities does the "Application Insights" system offer for web services, databases, and other cloud-deployed systems?

Application Insights provides:

- Detailed telemetry: requests, response times, dependencies, exceptions, availability.
- Smart alerts: automatic notifications when anomalies occur.
- Application Map: visualizes how services, APIs, and databases interact.
- Profiling and diagnostics: live profiler, failure analysis, distributed tracing.
- Dashboards and KQL queries: custom charts and deep data exploration.
- DevOps integration: CI/CD insights, deployment tracking, pipeline diagnostics.

– How might this functionality be beneficial in a corporate environment?

In a company Application Insights is key because it lets the monitor instantly like detecting and fixing app issues right away which cuts down on app downtime and protects the brand plus it helps save money because companies have data to use its Azure resources better and with that same information teams make decisions so the app and the user experience are always improving.

– How does the network layer contribute to the transmission of data between the web app and clients in different geographic locations?

The network layer, primarily utilizing the Internet Protocol, is instrumental in facilitating data transmission between the web application and geographically dispersed clients. Its core function involves logical addressing and dynamic routing, using source and destination IP addresses to determine the optimal path across the global network infrastructure so this complex process ensures that data packets are efficiently transported across various geographic regions with minimal latency, which is critical for maintaining high application responsiveness regardless of the client's location.

– In what ways do the application and transport layers ensure reliable data transfer and correct interpretation of web app responses?

The Transport Layer ensures reliable data transfer through sequencing, acknowledgment, and retransmission of packets, guaranteeing complete delivery. The Application Layer ensures correct interpretation by defining the structure of requests and responses, utilizing status codes to communicate the outcome of the transaction.

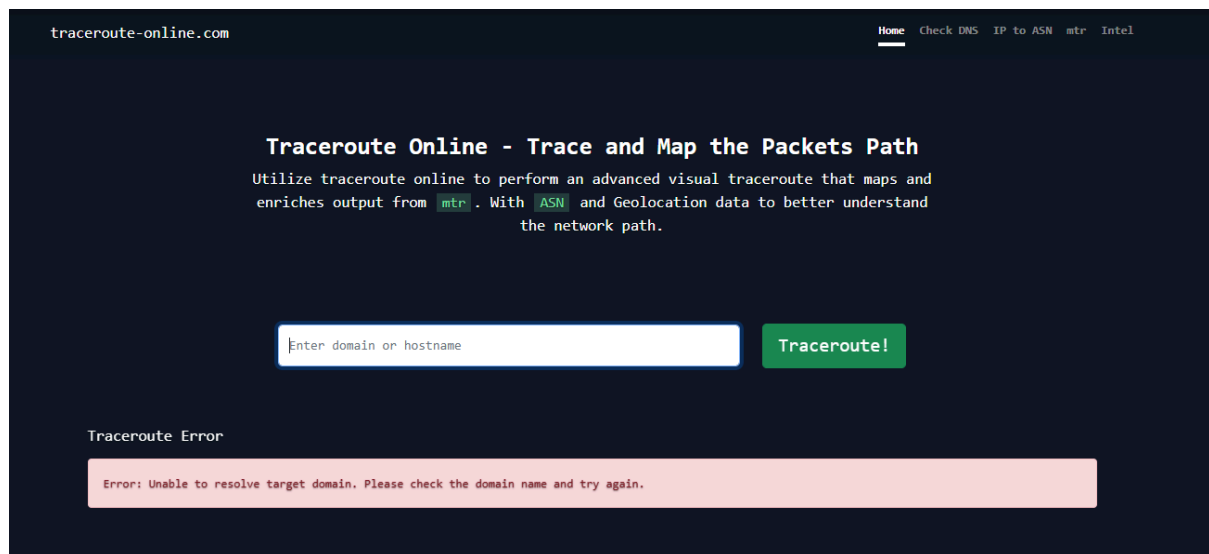
# Experiments

Perform the following tests in groups or individually, as instructed by the professor, and document the experience.

## 1. Use of ICMP Messages

a. Go to <https://tracertool-online.com/> and search for the Computer Science Laboratory page and the Stanford University page. Show the results.

We checked the page but it shows errors, we will try with any other options



This is the result for Stanford

| Query: ping stanford.edu, interval: 5s, dense mode: no                                       |               |      |      |        |        |        |        |       |       |       |
|----------------------------------------------------------------------------------------------|---------------|------|------|--------|--------|--------|--------|-------|-------|-------|
| Probing 171.67.215.288 - AS32 STANFORD - United States, California, Menlo Park [college] MAP |               |      |      |        |        |        |        |       |       |       |
| Link                                                                                         | ISP           | Loss | Send | Recv   | Avg    | Best   | Worst  | RTT   | Delay | Chart |
| Canada, BC, Vancouver                                                                        | Shaw          | 0%   | 14   | 33.81  | 39.96  | 33.65  | 49     | 13.74 | show  | link  |
| USA, CA, Fremont                                                                             | Hurricane     | 0%   | 14   | 2.31   | 2.24   | 1.88   | 2.72   | 0.2   | show  | link  |
| USA, CA, Fremont                                                                             | IT7 FMT2      | 0%   | 14   | 2.79   | 2.47   | 1.89   | 2.97   | 0.32  | show  | link  |
| USA, CA, Fremont                                                                             | Linode        | 0%   | 14   | 19.86  | 4.81   | 3.28   | 19.86  | 4.19  | show  | link  |
| USA, CA, San Francisco                                                                       | Digital Ocean | 0%   | 14   | 6.15   | 5.55   | 5.19   | 6.15   | 0.37  | show  | link  |
| USA, CA, Santa Clara                                                                         | Hurricane     | 0%   | 14   | 2.56   | 2.54   | 2.37   | 3.86   | 0.21  | show  | link  |
| USA, CA, Los Angeles                                                                         | Hurricane     | 0%   | 14   | 9.93   | 18.01  | 9.83   | 18.22  | 0.11  | show  | link  |
| USA, CA, Los Angeles                                                                         | DMIT          | 0%   | 14   | 9.54   | 18.88  | 9.51   | 18.42  | 0.41  | show  | link  |
| USA, CA, Los Angeles                                                                         | Vultr         | 0%   | 14   | 18.07  | 9.94   | 9.78   | 18.07  | 0.08  | show  | link  |
| USA, CA, Los Angeles                                                                         | NetActuate    | 0%   | 14   | 9.75   | 9.94   | 9.41   | 18.83  | 0.4   | show  | link  |
| USA, WA, Seattle                                                                             | Google        | 0%   | 14   | 19.25  | 19.65  | 19.2   | 28.65  | 0.49  | show  | link  |
| USA, UT, Salt Lake City                                                                      | Evolushost    | 0%   | 14   | 62.99  | 63.92  | 62.79  | 66.76  | 0.9   | show  | link  |
| USA, CO, Denver                                                                              | FDC Servers   | 0%   | 14   | 49.91  | 48.23  | 47.86  | 49.91  | 0.59  | show  | link  |
| USA, CO, Denver                                                                              | NetActuate    | 0%   | 14   | 27.12  | 27.72  | 27.85  | 28.59  | 0.44  | show  | link  |
| USA, TX, Fort Worth                                                                          | AT&T          | 0%   | 13   | 49.44  | 48.54  | 46.29  | 51.2   | 1.54  | show  | link  |
| USA, TX, Dallas                                                                              | NetActuate    | 0%   | 14   | 44.08  | 44.77  | 43.97  | 45.55  | 0.53  | show  | link  |
| USA, MO, Kansas City                                                                         | Rozint        | 0%   | 14   | 37.56  | 37.69  | 37.36  | 38.15  | 0.18  | show  | link  |
| USA, IL, Chicago                                                                             | NetActuate    | 0%   | 14   | 49.5   | 49.07  | 48.56  | 49.69  | 0.39  | show  | link  |
| USA, IL, Chicago                                                                             | Infraly       | 0%   | 14   | 49.72  | 49.62  | 49.11  | 50.08  | 0.99  | show  | link  |
| USA, IL, Chicago                                                                             | TeraCast      | 0%   | 14   | 51.94  | 52.31  | 51.73  | 53.44  | 0.6   | show  | link  |
| USA, FL, Miami                                                                               | Goodleaf      | 0%   | 14   | 75.84  | 76.71  | 75.5   | 87.37  | 2.98  | show  | link  |
| USA, FL, Miami                                                                               | TeraCast      | 0%   | 14   | 76.28  | 76.22  | 75.51  | 77.21  | 0.41  | show  | link  |
| USA, NY, Buffalo                                                                             | ColoCrossing  | 0%   | 14   | 78.97  | 71.42  | 78.66  | 72.12  | 0.48  | show  | link  |
| USA, NY, Buffalo                                                                             | Evolushost    | 0%   | 14   | 59.07  | 59.28  | 58.76  | 60.36  | 0.55  | show  | link  |
| USA, VA, Ashburn                                                                             | NetActuate    | 0%   | 14   | 64.15  | 64.82  | 63.51  | 64.58  | 0.31  | show  | link  |
| USA, NJ, Secaucus                                                                            | Goodleaf      | 0%   | 14   | 67.57  | 67.59  | 66.84  | 67.91  | 0.29  | show  | link  |
| USA, NY, New York                                                                            | Hurricane     | 0%   | 14   | 61.84  | 61.42  | 60.58  | 61.91  | 0.45  | show  | link  |
| USA, NY, New York                                                                            | TeraCast      | 0%   | 14   | 67.71  | 67.42  | 66.93  | 68.03  | 0.37  | show  | link  |
| Canada, QC, Montreal                                                                         | OVH           | 0%   | 14   | 64.52  | 64.44  | 64.05  | 65.08  | 0.3   | show  | link  |
| Chile, Santiago                                                                              | NetActuate    | 0%   | 14   | 161.41 | 161.56 | 161.08 | 163.02 | 0.51  | show  | link  |
| Paraguay, C.D.E.                                                                             | ASISPY        | 0%   | 14   | 188.91 | 189.28 | 188.69 | 190.08 | 0.41  | show  | link  |
| Brazil, Sao Paulo                                                                            | NetActuate    | 0%   | 14   | 178.42 | 179.38 | 169.72 | 171.14 | 0.48  | show  | link  |
| Brazil, Sao Paulo                                                                            | Hyperfilter   | 0%   | 14   | 173.03 | 172.18 | 171.29 | 173.15 | 0.56  | show  | link  |
| Morocco, Fez                                                                                 | Hostweb       | 0%   | 14   | 182.41 | 182.41 | 181.88 | 183.02 | 0.31  | show  | link  |
| Spain, Toledo                                                                                | PalasHost     | 7.1% | 14   | 247.27 | 206.11 | 201.87 | 247.27 | 11.89 | show  | link  |
| Spain, Madrid                                                                                | NetActuate    | 0%   | 14   | 132.85 | 133.37 | 132.68 | 134.11 | 0.46  | show  | link  |
| UK, Manchester                                                                               | NetActuate    | 0%   | 14   | 125.31 | 126.25 | 125.76 | 127.2  | 0.38  | show  | link  |
| UK, London                                                                                   | Cosmic Global | 0%   | 14   | 144.7  | 144.29 | 143.42 | 144.98 | 0.5   | show  | link  |
| UK, London                                                                                   | Evolushost    | 0%   | 14   | 128.29 | 127.78 | 127.88 | 128.43 | 0.47  | show  | link  |
| France, Paris                                                                                | Online.net    | 0%   | 14   | 135.95 | 136.56 | 135.55 | 137.61 | 0.47  | show  | link  |
| France, Paris                                                                                | NetActuate    | 0%   | 14   | 132.32 | 132.2  | 131.53 | 132.88 | 0.37  | show  | link  |

And this is the result for the computer science



Query: ping laboratorio.is.escuelaing.edu.co, interval: 5s, dense mode: no

Probing 45.239.88.88 - LACNIC /22 AS266862 ESCUELA COLOMBIANA - Colombia, Bogota D.C., Bogotá [business] MAP

| Geo                     | ISP           | Loss | Sent | Last   | Avg    | Best   | Worst  | StDev | MTR | Chart |
|-------------------------|---------------|------|------|--------|--------|--------|--------|-------|-----|-------|
| Canada, BC, Vancouver   | Shaw          | 0%   | 2    | 174.03 | 174.14 | 174.03 | 174.25 | 0.11  | ✓   |       |
| USA, CA, Fremont        | Hurricane     | 0%   | 2    | 118.15 | 118.1  | 118.05 | 118.15 | 0.05  | ✓   |       |
| USA, CA, Fremont        | IT7 FMT2      | 0%   | 2    | 112.38 | 112.64 | 112.38 | 112.89 | 0.26  | ✓   |       |
| USA, CA, Fremont        | Linode        | 0%   | 2    | 132.45 | 132.84 | 132.45 | 133.23 | 0.39  | ✓   |       |
| USA, CA, San Francisco  | Digital Ocean | 0%   | 2    | 117.7  | 117.97 | 117.7  | 118.24 | 0.27  | ✓   |       |
| USA, CA, Santa Clara    | Hurricane     | 0%   | 2    | 117.65 | 117.36 | 117.06 | 117.65 | 0.3   | ✓   |       |
| USA, CA, Los Angeles    | Hurricane     | 0%   | 2    | 106.87 | 106.32 | 106.07 | 106.58 | 0.26  | ✓   |       |
| USA, CA, Los Angeles    | DMIT          | 0%   | 2    | 118.7  | 118.81 | 118.7  | 118.92 | 0.11  | ✓   |       |
| USA, CA, Los Angeles    | Vultr         | 0%   | 2    | 97.99  | 97.81  | 97.63  | 97.99  | 0.18  | ✓   |       |
| USA, CA, Los Angeles    | NetActuate    | 0%   | 2    | 103.24 | 103.07 | 102.89 | 103.24 | 0.17  | ✓   |       |
| USA, WA, Seattle        | Google        | 0%   | 2    | 118.98 | 119.64 | 118.98 | 120.29 | 0.66  | ✓   |       |
| USA, UT, Salt Lake City | Evolushost    | 0%   | 2    | 129.57 | 129.7  | 129.57 | 129.82 | 0.13  | ✓   |       |
| USA, CO, Denver         | FDC Servers   | 0%   | 2    | 129.91 | 129.82 | 129.74 | 129.91 | 0.08  | ✓   |       |
| USA, CO, Denver         | NetActuate    | 0%   | 2    | 108.58 | 108.93 | 108.58 | 109.27 | 0.34  | ✓   |       |
| USA, TX, Fort Worth     | AT&T          | 0%   | 2    | 108.55 | 107.97 | 107.39 | 108.55 | 0.58  | ✓   |       |
| USA, TX, Dallas         | NetActuate    | 0%   | 2    | 91.51  | 91.58  | 91.51  | 91.64  | 0.06  | ✓   |       |
| USA, MO, Kansas City    | Rozint        | 0%   | 2    | 106.16 | 106.07 | 105.98 | 106.16 | 0.09  | ✓   |       |
| USA, IL, Chicago        | NetActuate    | 0%   | 2    | 101.41 | 101.14 | 100.87 | 101.41 | 0.27  | ✓   |       |
| USA, IL, Chicago        | Infraly       | 0%   | 2    | 97.38  | 102.11 | 97.38  | 106.84 | 4.73  | ✓   |       |
| USA, IL, Chicago        | Teracast      | 0%   | 2    | 102.67 | 103.19 | 102.67 | 103.7  | 0.52  | ✓   |       |
| USA, FL, Miami          | Goodleaf      | 0%   | 2    | 56.54  | 61.51  | 56.54  | 66.47  | 4.97  | ✓   |       |
| USA, FL, Miami          | Teracast      | 0%   | 2    | 61.94  | 61.78  | 61.61  | 61.94  | 0.16  | ✓   |       |
| USA, NY, Buffalo        | ColoCrossing  | 0%   | 2    | 98.94  | 99.56  | 98.94  | 100.17 | 0.62  | ✓   |       |
| USA, NY, Buffalo        | Evolushost    | 0%   | 2    | 104.72 | 104.64 | 104.56 | 104.72 | 0.08  | ✓   |       |
| USA, VA, Ashburn        | NetActuate    | 0%   | 2    | 95.9   | 95.85  | 95.79  | 95.9   | 0.05  | ✓   |       |
| USA, NJ, Secaucus       | Goodleaf      | 0%   | 2    | 106.46 | 106.67 | 106.46 | 106.88 | 0.21  | ✓   |       |
| USA, NY, New York       | Hurricane     | 0%   | 2    | 91.07  | 90.74  | 90.41  | 91.07  | 0.33  | ✓   |       |
| USA, NY, New York       | Teracast      | 0%   | 2    | 99.29  | 99.62  | 99.29  | 99.95  | 0.33  | ✓   |       |
| Canada, QC, Montreal    | OVH           | 0%   | 2    | 103.17 | 103.53 | 103.17 | 103.88 | 0.35  | ✓   |       |
| Chile, Santiago         | NetActuate    | 0%   | 2    | 126.94 | 127.22 | 126.94 | 127.49 | 0.27  | ✓   |       |
| Paraguay, C.D.E.        | ASISPY        | 0%   | 2    | 168.76 | 168.71 | 168.65 | 168.76 | 0.05  | ✓   |       |
| Brazil, Sao Paulo       | NetActuate    | 0%   | 2    | 151.51 | 151.08 | 150.66 | 151.51 | 0.42  | ✓   |       |
| Brazil, Sao Paulo       | HyperFilter   | 0%   | 2    | 152.85 | 152.5  | 152.15 | 152.85 | 0.35  | ✓   |       |
| Morocco, Fez            | Hostweb       | 0%   | 2    | 192.08 | 192.26 | 192.08 | 192.43 | 0.17  | ✓   |       |
| Spain, Toledo           | PalmaHost     | 0%   | 2    | 228.19 | 227.65 | 227.11 | 228.19 | 0.54  | ✓   |       |
| Spain, Madrid           | NetActuate    | 0%   | 2    | 181.55 | 181.21 | 180.87 | 181.55 | 0.34  | ✓   |       |
| UK, Manchester          | NetActuate    | 0%   | 2    | 170.88 | 170.74 | 170.6  | 170.88 | 0.14  | ✓   |       |
| UK, London              | Cosmic Global | 0%   | 2    | 174.83 | 174.84 | 174.83 | 174.85 | 0.01  | ✓   |       |
| UK, London              | Evolushost    | 0%   | 2    | 161.88 | 161.84 | 161.8  | 161.88 | 0.04  | ✓   |       |
| France, Paris           | Online.net    | 0%   | 2    | 157.8  | 157.66 | 157.51 | 157.8  | 0.15  | ✓   |       |
| France, Paris           | NetActuate    | 0%   | 2    | 170.7  | 170.7  | 170.7  | 170.7  | -     | ✓   |       |
| Netherlands, Amsterdam  | Online.net    | 0%   | 2    | 170.05 | 170.16 | 170.05 | 170.26 | 0.1   | ✓   |       |
| Netherlands, Amsterdam  | Hybula        | 0%   | 2    | 161.96 | 161.45 | 160.95 | 161.96 | 0.51  | ✓   |       |
| Netherlands, Amsterdam  | Interhost     | 0%   | 2    | 170.42 | 170.05 | 169.67 | 170.42 | 0.38  | ✓   |       |
| Netherlands, Eindhoven  | Evolushost    | 0%   | 2    | 176.99 | 177.62 | 176.99 | 178.24 | 0.63  | ✓   |       |
| Netherlands, Eindhoven  | YottaSrc      | 0%   | 2    | 171.99 | 171.94 | 171.88 | 171.99 | 0.06  | ✓   |       |
| Norway, Sandefjord      | Gigahost      | 0%   | 2    | 193.32 | 193.92 | 193.32 | 194.51 | 0.59  | ✓   |       |
| Switzerland, Zurich     | ifog          | 0%   | 2    | 175.59 | 178.02 | 175.59 | 180.44 | 2.42  | ✓   |       |
| Germany, Frankfurt      | ZetNet        | 0%   | 2    | 173.04 | 173.55 | 173.04 | 174.06 | 0.51  | ✓   |       |
| Germany, Frankfurt      | PalmaHost     | 0%   | 2    | 196.8  | 196.94 | 196.8  | 197.08 | 0.14  | ✓   |       |
| Germany, Nuremberg      | NetActuate    | 0%   | 2    | 175.71 | 175.6  | 175.48 | 175.71 | 0.12  | ✓   |       |
| Germany, Berlin         | Evolushost    | 0%   | 2    | 174.36 | 174.36 | 174.35 | 174.36 | 0.01  | ✓   |       |
| Slovenia, Maribor       | PipeHost      | 0%   | 2    | 185.32 | 184.66 | 184    | 185.32 | 0.66  | ✓   |       |
| Austria, Vienna         | PipeHost      | 0%   | 2    | 178.17 | 177.83 | 177.48 | 178.17 | 0.34  | ✓   |       |
| Austria, Vienna         | Evolushost    | 0%   | 2    | 170.37 | 170.58 | 170.37 | 170.79 | 0.21  | ✓   |       |
| Sweden, Stockholm       | Skhron        | 0%   | 2    | 194.46 | 195.06 | 194.46 | 195.66 | 0.6   | ✓   |       |
| Poland, Warsaw          | H2.Nexus      | 0%   | 2    | 211.64 | 211.29 | 210.94 | 211.64 | 0.35  | ✓   |       |
| Kosovo, Prishtine       | RS Computers  | 0%   | 2    | 203.33 | 203.01 | 202.69 | 203.33 | 0.32  | ✓   |       |
| Bulgaria, Sofia         | 7DC           | 0%   | 2    | 199.97 | 200.53 | 199.97 | 201.08 | 0.56  | ✓   |       |
| Greece, Athens          | Fusioned      | 0%   | 2    | 206.38 | 206.33 | 206.27 | 206.38 | 0.05  | ✓   |       |
| Finland, Helsinki       | H2.Nexus      | 0%   | 2    | 187.37 | 187.24 | 187.11 | 187.37 | 0.13  | ✓   |       |
| Romania, Bucharest      | ZetNet        | 0%   | 2    | 196.56 | 196.62 | 196.56 | 196.67 | 0.05  | ✓   |       |
| Romania, Bucharest      | M247          | 0%   | 2    | 201.16 | 200.67 | 200.18 | 201.16 | 0.49  | ✓   |       |
| Kyiv, Ukraine           | Starlink      | 0%   | 2    | 209.46 | 213.6  | 209.46 | 217.73 | 4.13  | ✓   |       |
| Turkey, Istanbul        | Oneprovider   | 0%   | 2    | 209.26 | 209.24 | 209.21 | 209.26 | 0.02  | ✓   |       |
| Turkey, Istanbul        | Teknosos      | 0%   | 2    | 224.89 | 225.06 | 224.89 | 225.22 | 0.17  | ✓   |       |
| Turkey, Istanbul        | YottaSrc      | 0%   | 2    | 203.13 | 202.72 | 202.31 | 203.13 | 0.41  | ✓   |       |
| Turkey, Bursa           | NetWeb        | 0%   | 2    | 223.35 | 223.22 | 223.09 | 223.35 | 0.13  | ✓   |       |
| Cyprus, Limassol        | CLS           | 0%   | 2    | 213.09 | 213.52 | 213.09 | 213.95 | 0.43  | ✓   |       |
| Israel, Tel Aviv        | Oneprovider   | 0%   | 2    | 227.25 | 227.97 | 227.25 | 228.69 | 0.72  | ✓   |       |

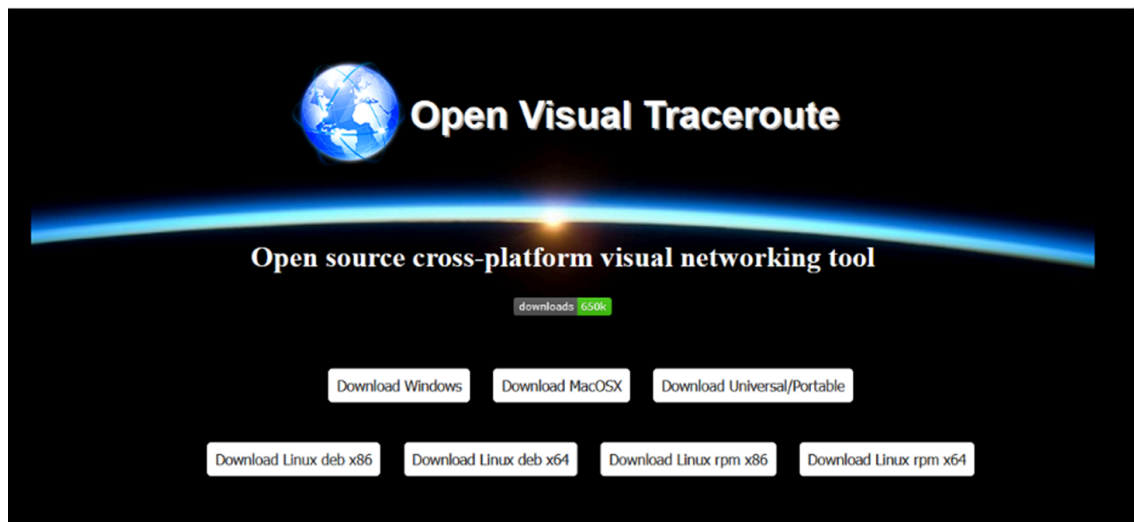
b. Using the tracer or traceroute command, search for a page in France and check the route.



For the second point, we performed a traceroute to a page located in France using the ping.pe tool, since the original page suggested in the lab had a 403 error. We used the domain [www.france.fr](http://www.france.fr)

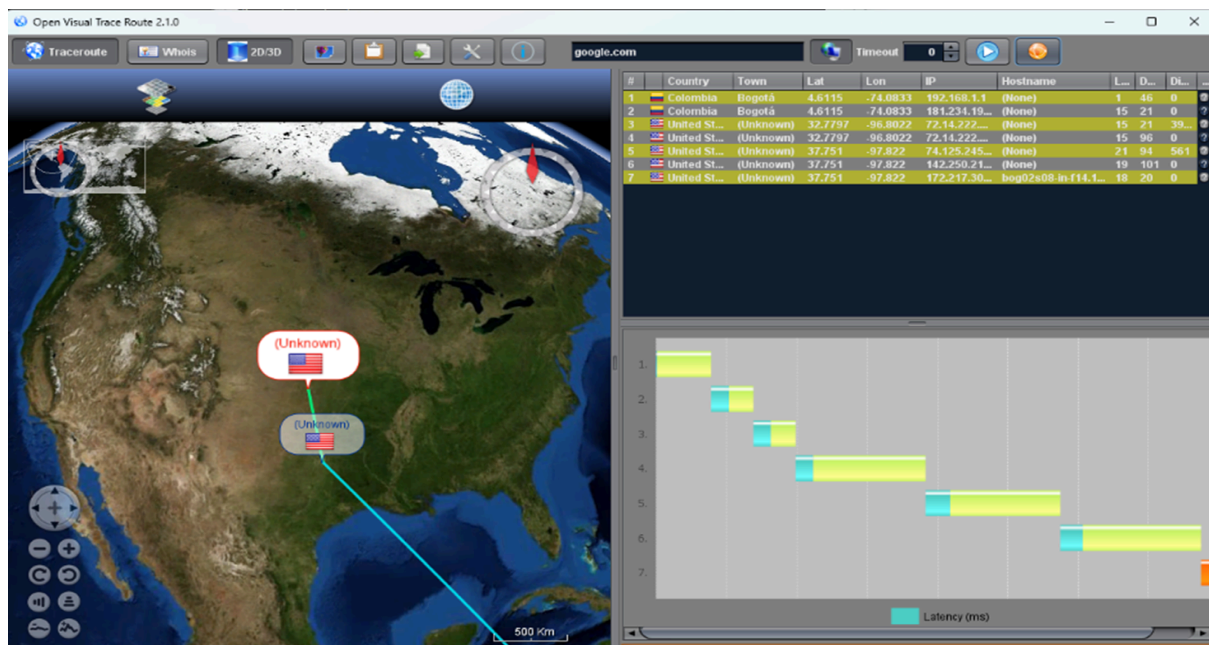
c. Download and install software such as VisualRoute, Open Visual Traceroute, or similar. They can be free tools or demos. Note: Only install this type of software, do not include other utilities or toolbars suggested during installation. Use advanced mode during installation to check what is being installed.

We installed Open visual Traceroute



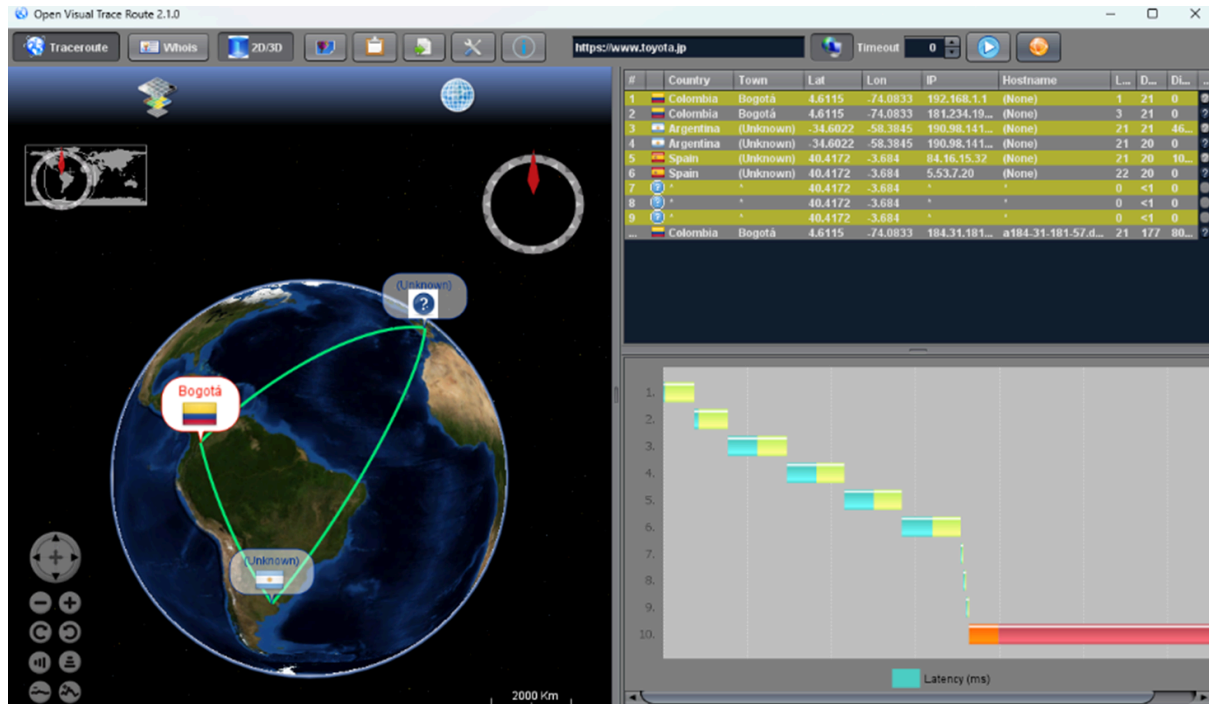
d. Test the tool, get to know some of its features.

We tested the application, we searched google.com, and the program showed us the hops made from where the request originated (Bogotá) to its destination. The program allows us to analyze these hops in a very visual and simple way.

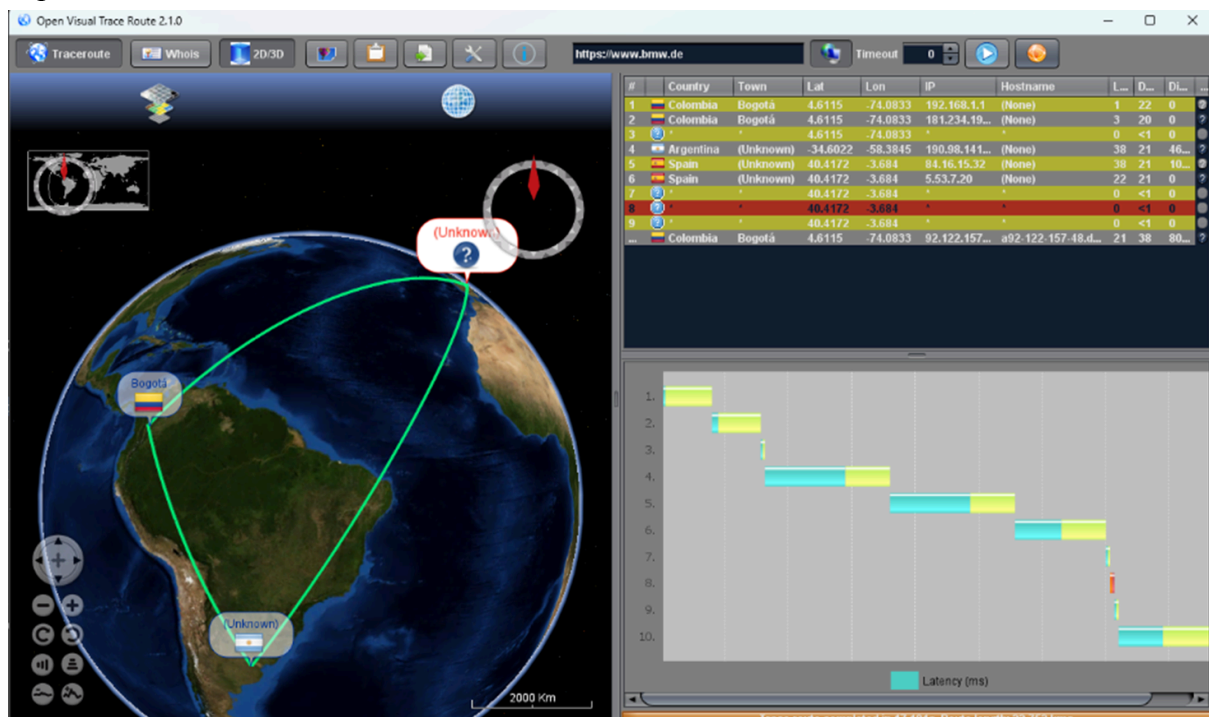


e. Document the tool's functionality by searching for 5 car manufacturers' websites around the world with the tool. Note: The websites should be located in different countries, preferably on different continents.

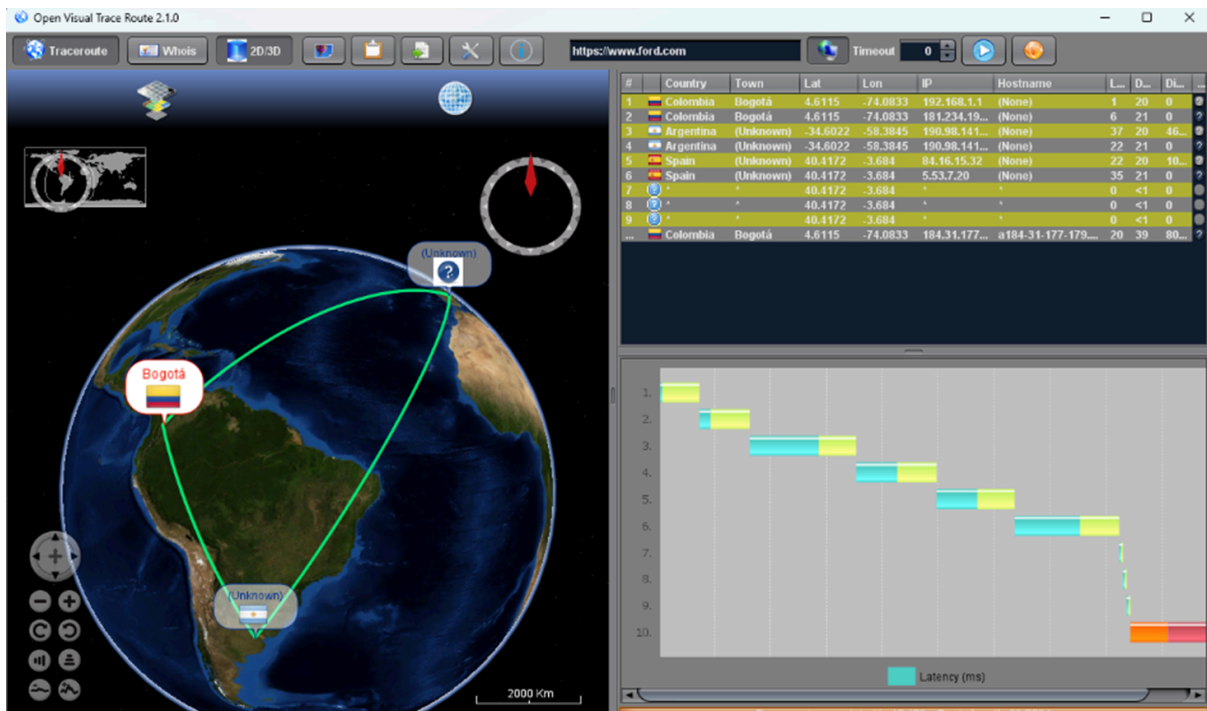
<https://www.toyota.jp>



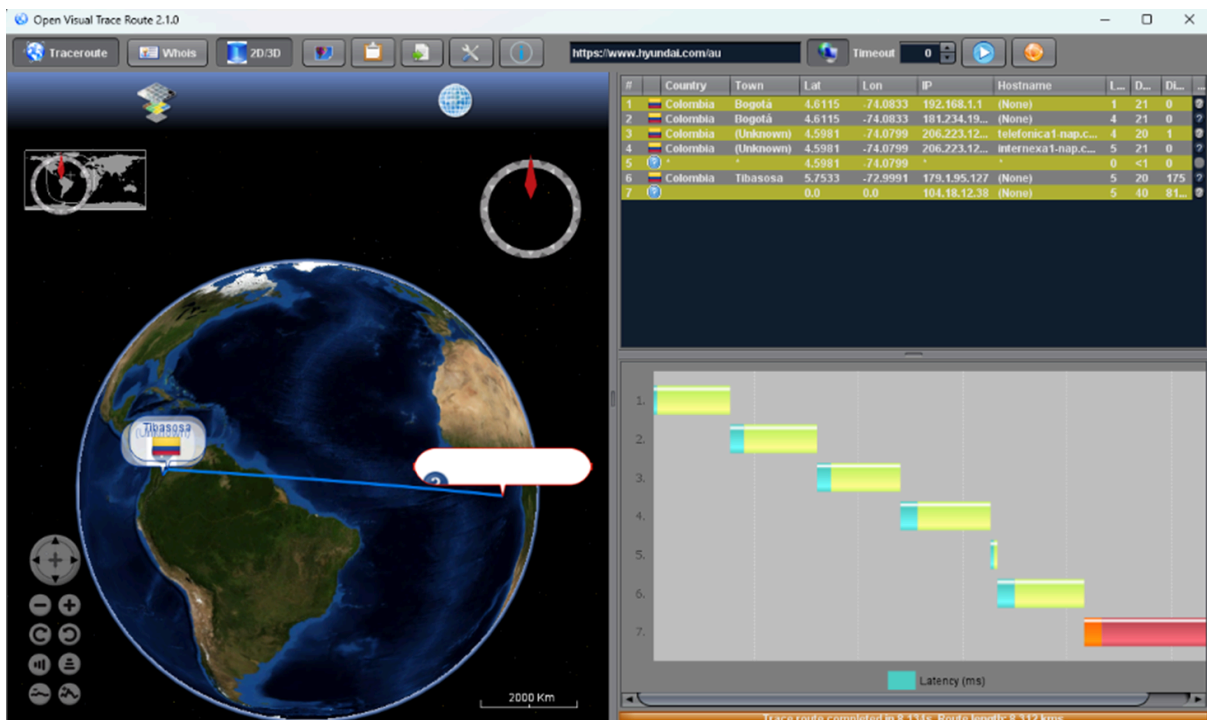
<https://www.bmw.de>



<https://www.ford.com>



<https://www.hyundai.com/au>



<https://www.renault.fr>



## 2. Some Questions About Router Commands

1. What is the difference between enable password and enable secret? If both are configured, which one takes precedence?

The command enable password stores the password using weak, easily reversible Type 7 encryption and enable secret uses strong, industry-standard MD5 hashing which is far more secure. If both are configured, enable secret always takes precedence and will be the required password for the user to enter privileged mode.

2. What is the difference between console and VTY?

The Console line provides physical and local access via a dedicated cable, used primarily for initial setup or recovery, supporting only one user at a time. VTY lines provide virtual, remote access over the network using protocols like Telnet or SSH, used for daily management and supporting multiple concurrent users.

3. What is the boot process of the routers in the Network Laboratory?

The router executes the post, then loads the Bootstrap from ROM, which locates and loads the full IOS image then Finally, the IOS copies the startup configuration from NVRAM into RAM to establish the running configuration

4. What types of memory do the routers in the Network Laboratory have?

ROM

Flash Memory

NVRAM

DRAM

5. What is the difference between the startup-configuration and running-configuration files?  
The startup-configuration is the saved configuration file stored permanently in NVRAM, while the running-configuration is the current, active configuration loaded into DRAM that controls the device's immediate operation.

### 3. Setup: Access and Basic Configuration of Routers

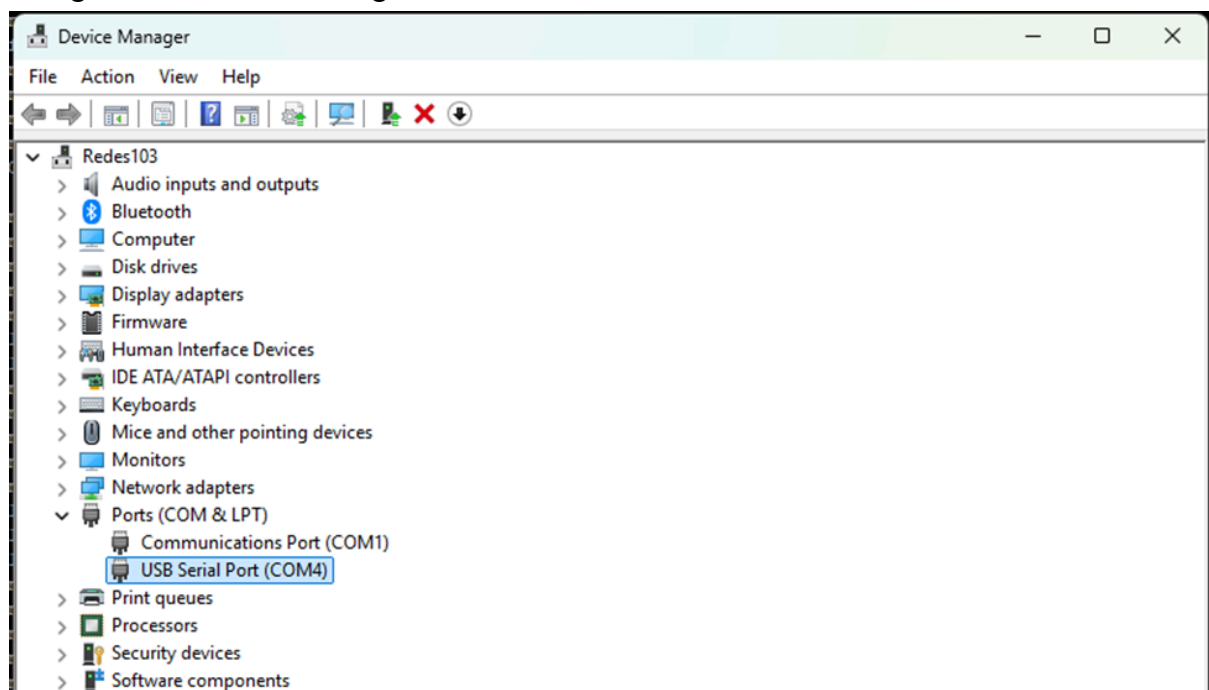
- Review and document the different routers available in the Laboratory and the network interfaces they have.

The networking lab has a number of Cisco routers which can be configured using a console cable between the router and the PC.

- To connect to the routers through the console, use a console cable and the hyperterminal or putty application, similar to how you connected to the switches.



- When starting the router, messages similar to those appearing on the switches or in the simulator (Packet Tracer) will be displayed. Try to log into the router and enter privileged mode. If successful, you can begin configuring the network indicated later. If access to the router or its privileged mode is password-protected, you will need to remove that configuration to start working with the device.







```
Router>enable
Router#
```

- For Cisco routers, explain the process the router goes through on boot when configured in modes 0x2142 and 0x2102.

- Perform the following configuration using physical devices and document the process. The basic configuration of a router should include:
  - Access keys for privileged mode, console, and remote access. The privileged mode key should be "cisco", the console key "claveC", and the remote access (telnet) key "claveT".
  - Router name. Assign the router the last name of one of the students in the group.
  - Console and remote access screen synchronization.
  - Description of the interfaces used.
  - Disable remote command server lookup. – Message of the day.

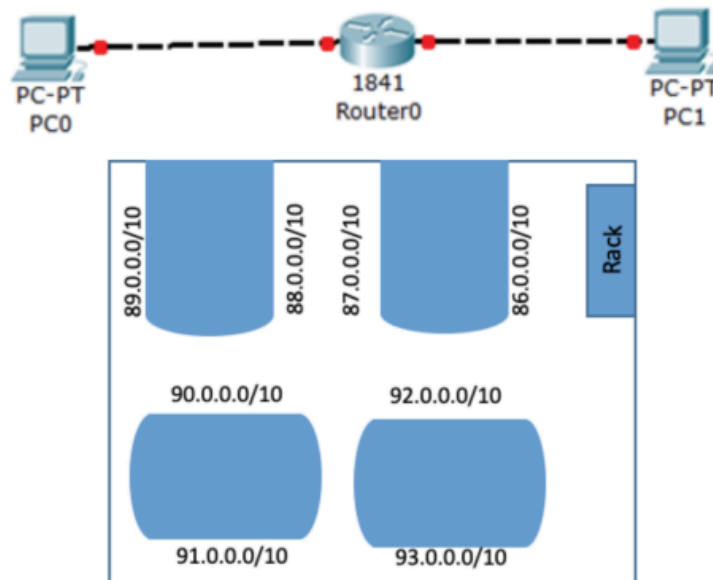
```
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#enable secret cisco
Router(config)#line console 0
Router(config-line)#passw
Router(config-line)#password claveC
Router(config-line)#login
Router(config-line)#logging synchronous
Router(config-line)#exit
Router(config)#line vty 0 4
Router(config-line)#password claveT
Router(config-line)#login
Router(config-line)#logging synchronous
Router(config-line)#exit
Router(config)#hostname amaya
```

```

amaya(config)#no ip domain-lookup
amaya(config)#banner motd # Only Access Authorized
Enter TEXT message. End with the character '#'.
banner motd #Only Access Authorized

```

- For the network on the left, use the range that allows addressing 4000 hosts, and for the network on the right, a range that allows addressing 600 hosts. Use the IP address range according to the location of the Network Laboratory desk diagram.



Recommendation: First configure the router through the console cable, then proceed with configuration and connection to the PCs.

- Test connectivity between computers.

```

Router(config)#interface GigabitEthernet 0/0
Router(config-if)#ip address 87.0.16.1 255.255.240.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface GigabitEthernet 0/1
Router(config-if)#ip address 87.0.4.1 255.255.252.0
Router(config-if)#no shutdown
Router(config-if)#exit

```

```
C:\Users\Redes>ping 87.0.4.1
```

```
Pinging 87.0.4.1 with 32 bytes of data:
```

```
Reply from 87.0.4.1: bytes=32 time<1ms TTL=255
```

```
Reply from 87.0.4.1: bytes=32 time<1ms TTL=255
```

```
Reply from 87.0.4.1: bytes=32 time<1ms TTL=255
```

```
Reply from 87.0.4.1: bytes=32 time<1ms TTL=255
```

```
Ping statistics for 87.0.4.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\Users\Redes>ping 87.0.16.2
```

```
Pinging 87.0.16.2 with 32 bytes of data:
```

```
Reply from 87.0.16.2: bytes=32 time=1ms TTL=127
```

```
Reply from 87.0.16.2: bytes=32 time=1ms TTL=127
```

```
Reply from 87.0.16.2: bytes=32 time=1ms TTL=127
```

```
Reply from 87.0.16.2: bytes=32 time=1ms TTL=127
```

```
Ping statistics for 87.0.16.2:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

```
C:\Users\Redes>
```

#### 4. Setup - Serial Interconnection

Join another group of students and interconnect the routers used in the previous step. For the interconnection, use the serial cards on the devices. For serial connections, agree on the network with your groupmates (do not use the same as other classmates—check with the professor). Use a subnet of the network 100.0.0.0/24



- Using the ping command, verify communication from router 0 and router 1. Note: Computer-to-computer interconnection will not work yet!

- What is a null modem?



A null modem is a type of serial cable used to directly connect two DTE (Data Terminal Equipment) devices without a modem in between.

It works by crossing the transmit and receive lines, allowing the devices to communicate directly.

In simple terms, it allows two computers or routers to exchange data using serial ports without needing modems.

- What is the clock rate command used for in routers, and why is it needed?

The clock rate command is used on the DCE side of a serial connection in Cisco routers.

Its purpose is to provide a timing signal (clocking) for the serial link.

In WAN serial links, one device must act as the timing source, and the other device must synchronize with it.

In the lab environment, since two routers are connected back-to-back with no real service provider, one of them must supply this timing signal.

That is why the clock rate command is required in lab setups.

- What does DTE and DCE mean? What is the relationship with the routers in the Network Laboratory?

- **DTE (Data Terminal Equipment):**

The device that receives the clocking signal.

Examples: computers, routers, PCs.

- **DCE (Data Communications Equipment):**

The device that provides the clocking signal.

Examples: modems, CSU/DSUs, and some serial cables in labs.

Relationship in the Network Laboratory:

In the networking lab, when two routers are connected using a serial DTE/DCE crossover cable, one end of the cable is marked DCE and the other DTE.

The router connected to the DCE end must configure the clock rate because it acts as the timing source.

The router connected to the DTE end simply receives the timing and does not set a clock rate.

This setup simulates the behavior of a real WAN provider inside the lab.

## 5. Static Routing

- Configure the routers with static routes so that you can ping all devices on the network.

| Router(config)# <b>ip route</b> network-address subnet-mask<br>{ip-address   exit-interface } |                                                                                                                                       |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Parameter                                                                                     | Description                                                                                                                           |
| network-address                                                                               | Destination network address of the remote network to be added to the routing table.                                                   |
| subnet-mask                                                                                   | Subnet mask of the remote network to be added to the routing table. The subnet mask can be modified to summarize a group of networks. |
| ip-address                                                                                    | Commonly referred to as the next-hop router's IP address.                                                                             |
| exit-interface                                                                                | Outgoing interface that is used to forward packets to the destination network.                                                        |

Para borrar la ruta: no ip route ....

Note: Each router must include routes (output interface) to reach networks that are not directly connected.

- Using the ICMP traceroute command, try to verify the communication between the two computers.
- Now, configure static routes to see the networks of your groupmates in the other group.
- Verify operation with ping and traceroute commands.
- Interconnect all course group networks and configure the routers so they can see all networks. Define IP ranges for the new serial connections between all.
- Verify operation with ping and traceroute commands.

```
Router(config)#interface GigabitEthernet 0/0
Router(config-if)#ip address 87.0.16.1 255.255.240.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface GigabitEthernet 0/1
Router(config-if)#ip address 87.0.4.1 255.255.252.0
Router(config-if)#no shutdown
Router(config-if)#exit
```

## 6. Closure

Erase and organize the routers and equipment used so that they are ready for a new practice session.

If I don't run copy running-config startup-config on a Cisco router, everything I configure stays only in the running configuration, which is stored in RAM and gets erased when the router restarts. The router only keeps what's saved in the startup configuration, so since I didn't save anything, all my changes disappear after a reboot. That means the router goes back to how it originally was, so I don't need to do anything else to leave it as it was.

## Conclusion

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Throughout this laboratory, we strengthened our understanding of how modern network infrastructures operate by working directly with operating systems, monitoring tools, and physical network devices. Implementing network-related shell scripts for Slackware, Solaris, and PowerShell allowed us to explore platform differences while applying POSIX-compliant techniques. This reinforced our ability to interpret system-level network information, manage ports and services, and create automated tools useful for troubleshooting in real environments.

Additionally, the installation of monitoring platforms using SNMP highlighted the importance of observability in corporate infrastructures. By enabling real-time monitoring of virtual machines and analyzing performance metrics, we experienced how organizations maintain visibility over CPU, memory, storage, and network behavior. The Azure exercise further showed how cloud-based services integrate telemetry tools like Application Insights to provide deep insight into application performance, user activity, and network layer behavior.

Finally, working with Cisco routers, serial connections, and static routing solidified our understanding of the core processes behind packet routing. Recovering router access, configuring interfaces, defining static routes, and analyzing ICMP traceroutes provided a practical foundation in network administration. Overall, the laboratory connected theoretical concepts with hands-on practice, giving us a comprehensive view of how local networks, cloud systems, and physical infrastructure work together in real-world environments.

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